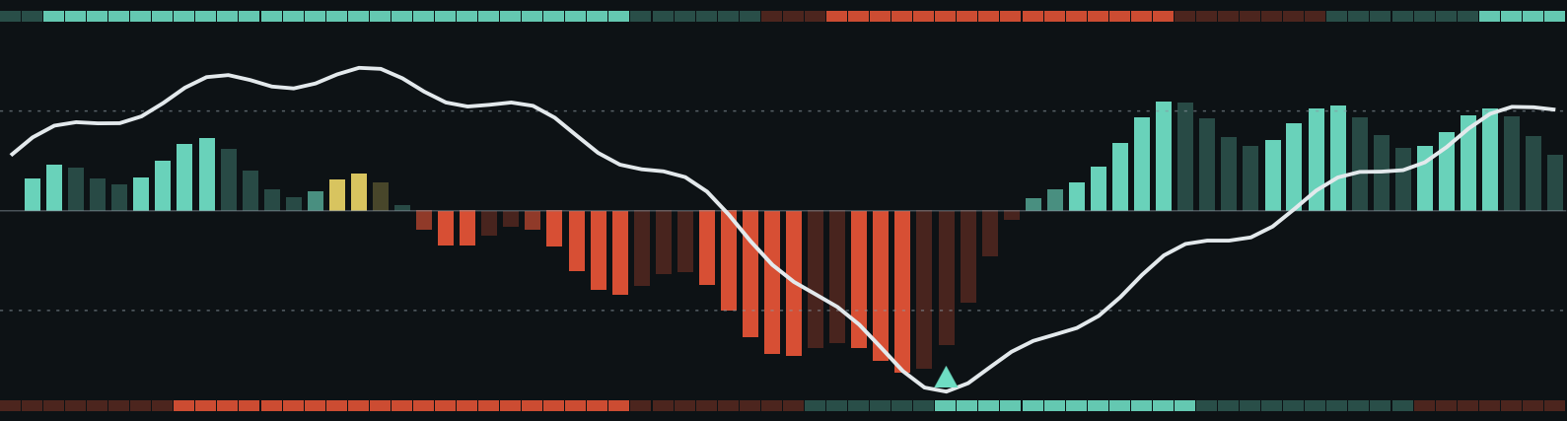
BULLISHVALUE

FOR PRAGATI V9.9 ON TRADINGVIEW

Pragati

Field Manual

"Is this move paid for, and is it paid for on every timeframe?"



READ

Every element of the pane and the price chart, and every state of the readout.

INVEST

The Conviction-Value Grid as an allocation map, with the capitulation plan.

TRADE

Accepted momentum, the system that passed a held-out test, with its exact rules.

Manual 2.0 · November 2026

Educational, not financial advice

Pragati

Field Manual

"Is this move paid for, and is it paid for on every timeframe?"
How to read the indicator, invest with its grid and trade its tested trend system.

BULLISHVALUE

Pragati Field Manual
Manual 2.0 · November 2026
For Pragati v9.9 on TradingView (branch `pragati-v9.9`)

Published by BullishValue. © 2026 BullishValue. The Pragati script is released under the Mozilla Public License 2.0.

Not financial advice. This manual is educational. It describes how an indicator works and how it was tested. Past results, including results on held-out data, do not guarantee future results. Trade only what you can afford to lose, and paper-trade first.

Sources. Measured results are from `docs/trail-lab.md` and `docs/trail-tradability.md` in the Pragati repository. Results marked External are quoted from the v9 audit, which lives in a separate repository and was not re-checked for this edition.

Set in IBM Plex Serif, IBM Plex Sans, IBM Plex Mono and Archivo.

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How to use this manual

Pragati answers one question about a market: whether a move is paid for, and on every timeframe. This book teaches it in the order a reader becomes fluent. Part I is the foundation and Part II the knowledge base, every engine from the bar up. Part III reads every state; Part IV reads them in context; Part V separates fakeouts from acceptance, with measured numbers. Parts VI and VII are the trade and investment guides, Part VIII six real markets reconstructed, Part IX the combinations, Part X drills with answers, and Part XI the reference.

Read Part I first, then follow the mastery path in chapter 3. Each chapter opens with what it will let you do.

How well each claim is known

Pragati was built and audited in public, and many of its parts were tested on markets, but not all of them. Every claim carries one of three tags.

MEASURED

Tested in this repository on real market data with a pre-registered plan, and on data the design never saw. The numbers are in `docs/trail-lab.md` and `docs/trail-tradability.md`.

EXTERNAL

Measured in the v9 audit (380 instruments, six asset classes, three eras), whose data lives in a separate repository. The numbers are quoted in the script and were not re-checked here.

UNMEASURED

A reading aid or a reasoned combination. Useful for context, never proven to make money. Size accordingly.

Conventions

- Text in a monospace face, and coloured chips such as `LONG 0.62`, show what the indicator itself prints.
- "Section 16.2" refers to a numbered section, and "chapter 11" to a chapter that has no sections. Figures and tables are numbered by chapter.
- Case studies marked *reconstructed* are real daily bars run through the repository's reference engine; walkthroughs marked *constructed* were written to teach, and no market produced them.
- ▲ ▼ ◆ × are the marks the indicator draws. Gold always means caution and never a direction.
- Every figure in this manual is drawn from the indicator's defaults on daily charts, which is how it was tested.

PART I

Foundations

1	Getting started	3
2	The visual language	5
3	How Pragati thinks	8

CHAPTER 1

Getting started

1.1 Setup in five minutes

1. **Add Pragati to the chart.**
   It opens in its own pane under price, and it draws the trail, the volume profile and the readout table on the price chart itself.

2. **Use standard candles.**
   On Heikin Ashi, Renko, Kagi, Point & Figure or range bars the prices never traded, so every family pauses and says "non-standard chart".

3. **Start on the daily chart.**
   Daily bars are what was tested. Weekly bars were tested for the ▲ and the grid. Intraday works, but nothing on it was measured.

4. **Leave the defaults alone at first.**
   View `Full`, Trace `Conviction × Value`, conviction on `Ladder down`, value on `Ladder up`, Accepted momentum `on`. Every number in this manual was measured at these settings.

5. **Give it history.**
   Conviction's histogram needs about 450 bars to calibrate (1.8 years of daily bars). Until then the push gate is warming: the grid's rows have no memory, the trail cannot start, and the readout says so.

6. **Create one alert.**
   Choose the condition "Any alert() function call" on Pragati. It carries every event once, at its bar's close (section 31.1).

1.2 Reading a chart in sixty seconds

Five looks, in this order, at the close of a bar. The worked readings come from the readout in Figure 2.2.

1. **Is there a reading?**
   Anything paused, warming, stale or gold means smaller or later. In Figure 2.2 nothing is gold.

2. **Trader: the AUCTION row's CHART cell.**
   `LONG 0.62`: the tested system is long, at size 0.62. Size it with the formula in section 18.1.

3. **Investor: the CVG row's STATE.**
   `Buy 2.71`: the TURNED family (buyers in control, price cheap). Weight it at 2.71 units in the book.

4. **Why: the CVG row's WHY and PUSH.**
   "buyers running · turning up" and ↑: both edges are confirmed and a push is on. Nothing is held.

5. **Context: OI and the auction.**
   "Long build-up · book 62 pct": new longs, the book not crowded. "Migrating ↑ · trail ↑ · in value": value is moving with the trail. A stop belongs below the trail or the session VAL.

CHAPTER 2

The visual language

2.1 The colour grammar: one hue, one job

Colours mean the same thing everywhere in Pragati. Learn these six and every element reads itself.

Green · direction up, action buy
Buyers in control, a rich value reading, a long position, a Buy or Accumulate action.

Red · direction down, action sell
Sellers in control, a cheap value reading, a short position, a Trim or Exit action.

Gold · caution, never a direction
Read with care: a probe, a quiet regime, a held row, a crowded book, short covering, crypto's capitulation.

Grey · neutral
Nothing to read, Hold or Wait, Accepted momentum flat.

Blue · the session profile's value area
Identity, not direction. The VAH, VAL and value rows.

Slate · the trail's rolling value ribbon
Identity: a different value area from the session profile's.

> On the value tape, green means **rich** and red means **cheap**. A red value tape is a price below fair value, which is what the capitulation ▲ wants.

2.2 The pane, labelled

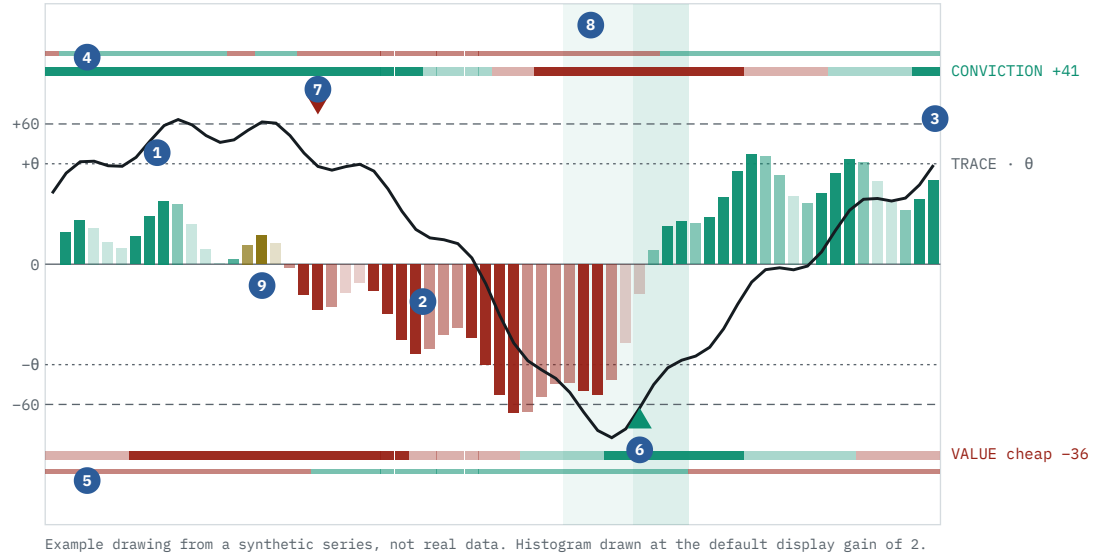
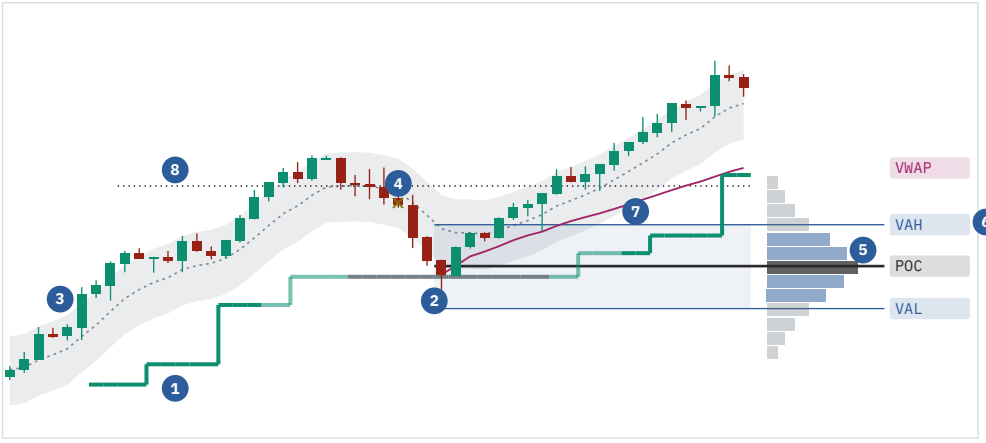


Figure 2.1 The oscillator pane. Numbers match the list below.

1. **Trace.** Default Conviction × Value: how stretched the move is, in effort and in price against fair value. Rails at ±θ (±43 at the default) and ±60, a 3σ stretch. UNMEASURED
2. **Histogram.** The trace minus its 9-bar signal. Hue is the direction of the push. Bright means pressing, mid means decelerating, pale means turning. UNMEASURED
3. **Rails.** Dotted ±θ, dashed ±60. On the Conviction trace they are the inner (±30) and outer (±60) zones.
4. **Conviction tape (+82)** with its momentum band (+89). The ladder's view of who is in control: green buyers, red sellers, bright past the inner zone.
5. **Value tape (−82)** with its momentum band (−89). Samanvaya's multi-timeframe value: green rich, red cheap, bright past θ.
6. ▲ **Capitulation turn.** The grid stands in DOWN · cheap and value momentum has turned back toward fair. EXTERNAL
7. ▼ **Distribution.** Sellers have taken control of a price rich past θ. EXTERNAL
8. **Background tint.** Green while the grid holds Buy · capitulation, deeper after value momentum turns. Red in Exit · distribution. Gold on crypto.
9. **Gold cast.** Columns turn gold when the push was paid for more by exits than new positions, from open interest. EXTERNAL

A column's colour is not who is in control. Control is the trace's side of zero. A green column under a trace at −45 means sellers still control, with their grip easing. When the market is unusually quiet the columns lose half their colour: a loud pane in a quiet market is drawn quiet.

2.3 The price chart, labelled



	STATE	WHY	PUSH	CHART	SPAN
OI	Long build-up	new longs · book 62 pct	+1.4σ		3d
CVG	Buy 2.71	buyers running · turning up	↑	same	6d
AUCTION	Migrating ↑	trail ↑ · in value	↑	LONG 0.62	41d

9

Example drawing from a synthetic series, not real data. The readout is the default bottom-left table.

Figure 2.2 On the price chart. Numbers match the list below.

1. **Accepted momentum trail.** The tested engine. Hue is its side, intensity its size, grey when flat. MEASURED
2. **Flat segment.** The trail's side and momentum disagree, so the position is zero.
3. **Value ribbon** (slate) with its dotted POC. Where 70% of the trail window's participation traded.
4. **Probe ×** (gold). The first close beyond the trail that the market did not accept.
5. **Session volume profile.** Rows by price: the POC row bright, value-area rows blue, tails grey. Session = day intraday, week on daily, month on weekly. UNMEASURED
6. **Level tags.** POC, VAH, VAL, VWAP (TWAP when volume is off), IBH and IBL intraday, prior-session levels.
7. **Session VWAP** (rose).
8. **Naked POC.** A prior session's POC that price has not traded back through since. Up to 5, up to 250 bars old.
9. **The readout.** Three rows (OI, CVG, AUCTION) and five columns (STATE, WHY, PUSH, CHART, SPAN). See chapter 12.

CHAPTER 3

How Pragati thinks

3.1 The one question

> **AFTER THIS CHAPTER YOU CAN**
>
> - say in one sentence what Pragati measures, and in one more what it does not claim

"Is this move paid for, and is it paid for on every timeframe?" Everything in Pragati is a way of asking that question, and each engine asks a different part of it:

Paid for in effort
Conviction: how much of the bars' travel became displacement, and how consistently, weighted by participation. A move that went far on little agreement is not paid for.

Paid for in price
Value: where price stands against a model of fair. A move that has already carried price rich is paid for in the other sense: it is late.

Paid for in acceptance
The auction: whether the market's value has followed price. A break that value has not followed is a probe.

On every timeframe
The ladder reads conviction from inside the bar and value from above it, so one chart answers on three horizons.

What it does not claim. Most of what Pragati draws *describes*. Measured without look-ahead, the pane's readings carry almost no timing information of their own `EXTERNAL`. Three things here carry measured leans:

- the grid's capitulation state and the ▲ that marks its turn `EXTERNAL`;
- the grid's graded units across a book `EXTERNAL`;
- Accepted momentum, across many markets `MEASURED`.

The setup lab adds measured facts about breaks: what a probe does to a trend, and what gets accepted `MEASURED`. Everything else is structure for reading, tagged `UNMEASURED`.

Design rules you will meet everywhere:

- Gold means caution, never direction.
- Nothing pauses silently.

- Nothing that fires repaints.
- Every claim carries its evidence.

3.2 The engines in plain words

Conviction

How much of the window's effort turned into net movement, weighted by participation (volume, or true range where there is none).

Answers: who is in control? · Seen in: the Conviction trace, the conviction tape, the histogram.

Value (Samanvaya)

Where price stands against fair value: a macro-hedged relative-value leg blended with a breadth leg, in σ. Past θ is cheap or rich.

Answers: is the price stretched? · Seen in: the value tape, the Value trace.

The ladder

Both readings across timeframes. Conviction on Ladder down reads the bars inside each chart bar. Value on Ladder up reads the chart and the frames above.

Answers: is it true on every timeframe? · Seen in: the two tapes.

The Conviction-Value Grid

The two tapes as a 3 × 3 map. Each cell is a family with an action and units, the weights Pragyam sizes from. EXTERNAL

Answers: what should a portfolio hold? · Seen in: the CVG row, the ▲▼, the tint.

The auction

A rolling volume profile of the chart's own bars: where participation was accepted, and whether value is migrating.

Answers: where has the market agreed on price? · Seen in: the trail, the ribbon, the AUCTION row.

Accepted momentum

The six-month acceptance trail sets the side; time-series momentum sets the size. Long only when both say up, short only when both say down. MEASURED

Answers: should a trend book be long, flat or short? · Seen in: the trail's colour, the AUCTION row's CHART cell.

Open interest

On futures: whether this bar's trading opened risk (builds) or closed it (covering, unwinding). EXTERNAL

Answers: what kind of positioning? · Seen in: the OI row, the gold cast.

Session volume profile

Where volume traded in the current session: POC, value area, VWAP, initial balance, naked POCs. UNMEASURED

Answers: where are the reference levels? · Seen in: the right edge of the price chart.

3.3 The mastery path

AFTER THIS CHAPTER YOU CAN

- plan how to work through this book

Table 3.1 Six levels

Level	You can	Read	Practise
1 · Reader	Name every mark and cell on the chart	Parts I and III	Drills 1–8
2 · Mechanic	Say which engine made a reading, from what, and when it is final	Part II	Rebuild a readout by hand from its rules

Level	You can	Read	Practise
3 · Contextual reader	Read a chart top-down and name its regime	Part IV	Drills 9–12; the five-minute routine on ten charts a day for two weeks
4 · Skeptic	Spot a trap, and score a break	Part V	Drills 13–18
5 · Practitioner	Run a setup card, size it and journal it	Parts VI and VII, with the cases in Part VIII	Drills 19–24; paper trading for at least six months
6 · Master	Know where the evidence stops, and act accordingly	Parts IX and XI	Teach the context stack to someone else

> Work the drills before reading the answers. Chapter 30's answers explain the reasoning and point back to the section that teaches it.

PART II

The knowledge base

How every engine works

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CHAPTER 4

The conviction engine

4.1 From a bar to a share

> **AFTER THIS CHAPTER YOU CAN**
>
> - compute a bar's conviction and its participation weight by hand
> - say what the Agreement and Effort denominators measure, and why the default is Agreement
> - explain why participation is capped

Conviction is Pragati's first instrument and its oldest. It asks how much of a bar's travel became displacement, and how consistently bars resolve one way, weighted by how much participation stood behind them.

One bar

```
c   = (close − previous close) ÷ true range          bounded −1 … +1
w   = min( volume ÷ EMA20(volume), 3 )             the participation weight
```

A bar that closes at its high after opening at the previous close has $c = +1$: every tick of its range became net movement. A bar that travels widely and closes where it started has $c \approx 0$, however violent it was.

The weight w is the bar's volume against its own 20-bar average. Where a bar prints no volume, the last good print is carried through hollow bars. Where the symbol has no volume at all, as with index spot, relative true range stands in. That is the `Auto` setting of `Participation`. `Volume`, `True range` and `Off` force one choice. Off gives equal weights, a control: if the reading barely changes with it off, volume tells you nothing on that symbol.

The cap at 3 matters more than it looks. Expiry days, index rebalances and block prints carry volume many times normal. Uncapped, one such bar would own the whole window for its full length, and the oscillator would be reporting that one bar. The original author's tests found participation weighting earns its place, with Off the worst setting in all three signal architectures tested EXTERNAL.

The window: the raw share

```
Agreement (default)   raw = 100 × Σ(c·w) ÷ Σ(|c|·w)     over the Lookback (20 bars)
Effort                raw = 100 × Σ(c·w) ÷ Σ w
```

Agreement is Chande's form, the RSI family, applied to range-normalised closes weighted by participation. It asks how *consistently* bars resolve one way. A bar that went nowhere drops out of both sums, so heavy volume with no progress is invisible to it. Every number in the script's evidence was measured on this form.

Effort divides by total participation, which is Wyckoff's law taken literally. A heavy bar that went nowhere dilutes the reading, as Chaikin money flow divides by volume. It is untested UNMEASURED. Whichever is chosen, the panel's `effort → result` row measures absorption under both (section 4.3).

Table 4.1 A worked raw share over four bars (constructed arithmetic)

Bar	Close change	True range	c	Volume ÷ average	w	c · w	|c| · w
1	+12	20	+0.60	1.2	1.2	+0.72	0.72
2	+3	15	+0.20	0.8	0.8	+0.16	0.16
3	−8	16	−0.50	2.0	2.0	−1.00	1.00
4	+10	12	+0.83	4.5	3.0 (capped)	+2.50	2.50

Over these four bars Σ(c·w) = +2.38 and Σ(|c|·w) = 4.38, so Agreement reads 100 × 2.38 ÷ 4.38 = **+54**: buyers won most of the weighted travel. Effort divides by Σw = 7.0 and reads **+34**. Bar 3's heavy, failed sale counts against buyers in both. Bar 4's spike counts as 3, not 4.5.

4.2 Scaling, zones and the quiet regime

AFTER THIS CHAPTER YOU CAN

- say what +30 and +60 mean on any instrument
- explain why a dead market can read loud, and how Pragati marks it

```
conviction = 100 · tanh( raw ÷ 3σ )        σ = standard deviation of raw over the Normalizat
```

The raw share is divided by three of its own standard deviations and squashed by tanh, so the reading is bounded at ±100 and calibrated to the instrument. The inner zone, **±30**, is occupied about a third of the time. The outer zone, **±60**, is occupied about 4% of the time: measured at 3.2% to 4.3% EXTERNAL.

Two consequences follow, and both are easy to forget.

- **Scaling is relative.** +60 is extreme *for this instrument over this window*. It is not a universal overbought line. Past the outer zone is where trends run, and also where they end.
- **Scaled, not centred.** Zero keeps meaning "neither side in control". On a persistently trending instrument one side's zone is simply visited more often.

The quiet regime

When σ itself collapses, a small imbalance divided by a small σ reads large. Pragati watches for this. The quiet regime is σ in the **bottom fifth of its own history**, ranked over four Normalization windows (800 bars at the defaults). While it holds:

- the histogram's columns lose about half their colour, while the trace carries conviction;
- the panel's raw-share row reads QUIET;
- conviction's **push gate shuts**. The grid's rows hold, and the acceptance trail cannot step (section 8.2).

Quiet is not rare, and it clusters. On the repository's reference code it held on about a fifth of all bars, as designed, but on **85% of NIFTY 50's last 300 bars** to October 2026 and 58% of gold's MEASURED. Chapter 25 shows what that does to the trail.

4.3 The histogram, its tiers and the push gate

> **AFTER THIS CHAPTER YOU CAN**
>
> - read a column's hue, lightness and saturation as three separate answers
> - name the four tiers and say which ones open the push gate
> - say how long the gate takes to warm up

The trace is smoothed by a short EMA (`Smoothing`, 3) and compared with its own signal line (`Signal EMA`, 9). The histogram is the gap:

```
histogram = trace − EMA9(trace)        impulse when |histogram| ≥ k · σ(histogram)    k = 0.
```

A plain crossing of trace and signal is **not** a signal here. Two lines near zero cross endlessly, and k = 0 (a plain crossover) was the worst value out of sample in a 900-configuration search EXTERNAL.

Table 4.2 The histogram's four tiers

Tier	Condition	Drawn	Push gate
0 · impulse	Growing away from zero, past k·σ	Near-solid	Open, on its side
1 · building	Growing away from zero, below k·σ	Moderate	Open, on its side
2 · decelerating	Shrinking, the trace still moving with it (by more than 0.1σ of its bar-to-bar change)	Mid	Open, on its side
3 · turning	Shrinking with the trace flat or against it	Pale	**Shut**

Hue is not control. A column's hue is which side of its signal line the trace is on, which is the direction of the *push*. A green column with the trace at −45 means sellers are still in control but their grip is easing.

The push gate

The gate is the one thing every layer of Pragati that "moves" waits for. The grid's rows move only on it (section 7.1), and the acceptance trail starts and steps only on it (section 8.2):

```
gate = +1   histogram ≥ 0, tier 0, 1 or 2, not quiet
gate = −1   histogram < 0, tier 0, 1 or 2, not quiet
gate =  0   tier 3 (turning), or the quiet regime
```

It always reads **conviction's own histogram**, whichever trace the pane draws. Only on the Conviction trace are the pane's columns the gate's columns.

Warm-up. The histogram's σ needs a clean 200-bar window of a histogram that itself needs the trace's 200-bar σ. The gate therefore reads from about **450 bars**, 1.8 years of daily bars. Until then the grid's rows have no memory, the trail cannot start, and the readout says `gate warming`.

Effort → result

The panel's row Σ|c|·w ÷ Σw is the share of the window's participation that became directional displacement. At 100%, every bar closed at its extreme. In the bottom fifth of its own history the row turns gold and reads `ABSORBED`: effort met by opposing orders, which can resolve either way. Only the ◆ reads it; requiring absorption did not improve the ▲ EXTERNAL.

CHAPTER 5

The value engine

5.1 Samanvaya's two legs

AFTER THIS CHAPTER YOU CAN

- say what 'cheap' and 'rich' mean in Pragati, and what they do not mean
- name the two legs and what each one sees

Value is Samanvaya's engine, carried into Pragati line for line. Where conviction asks *who is pushing*, value asks *where price stands against fair*. Cheap and rich are statistical statements about this price against a model of it. They are not statements about a company's worth.

Leg 1 · Relative value (RV)

The chart's own return is hedged against a stepwise-selected basket of macro drivers (rates, currencies, commodities and the like). A custom ninth driver can be added. The residual spread is scored over an ensemble of timescales.

The hedge is applied in proportion to its measured skill, its rolling out-of-sample R^2 (`Macro hedge · Auto`). The model is fitted on an anchored clock: one observation per closed day by default, or per week. It spends 22 of Pine's 40 data requests.

Leg 2 · Breadth

Seven causal views of the chart's own price: the Market Strength Factor at five timescales, plus the fastest as momentum only and the slowest as flow only. It is the price-only half of Tattva's Swayam ensemble.

Breadth is momentum, so it overlaps conviction. The panel's `layer overlap` row measures by how much rather than assuming it away.

The two legs are blended in z-space, with their correlation measured, at `Leg mix` 0.5: the equal blend Samanvaya measured best (+0.0691 against +0.0637 for RV alone) EXTERNAL. Set the mix to 1.0 to see value without momentum.

On every timeframe

The model takes one observation per closed anchor period, a day by default, and each observation is the **sum** of the chart-bar returns inside it. Before the fix, on a 5-minute chart the whole macro fit ran on one closing 5-minute return a day, and value carried no macro information. The panel's `value · macro share` row shows how much of value is relative value on this chart.

Driver timing. On a daily NSE chart many drivers close hours later; TVC:US10Y closes about eleven hours after NSE. A driver whose bar closes more than a third of a bar late is

read at its previous close, the value known at the chart's close. The panel's `value · driver timing` row shows what was lagged.

5.2 From z to θ: the value scale

> **AFTER THIS CHAPTER YOU CAN**
>
> - convert θ into the pane's units
> - say when the value tape leaves grey

```
value = 100 · softbound( 0.5 · z )          z = Samanvaya's unified z (RV ⊕ breadth)
θ     = 1.5 (in |z|)    →    ±43 on the pane and the tapes
```

Value is positive when price is **rich** and negative when it is **cheap**. θ, `θ · value threshold`, is where Samanvaya calls price cheap or rich: 1.5σ at the default, which lands at ±43. The literature standard for a mean-reversion entry is 1.5 to 2.5. θ sets:

- the grid's columns, and with them the ▲ (DOWN · cheap) and the ▼ (DOWN · rich);
- the value tape's knee and the ◆'s and H's value tests;
- the inner rail of the Value and blended traces.

Colours are flipped on purpose. On Pragati's value tape, green is rich and red is cheap, the reverse of Samanvaya's own pane. A rally makes price rich *and* conviction positive, so both tapes turn green together. A split in hue between them is then the thing the eye catches.

Warm-up. The value tape stays neutral grey until Samanvaya's model is built: ten anchor periods past a twelve-period hold-out, and the basket settled. While a basket settles after a rotation, a blended trace dims.

Measured. The 1.5σ entry and the leg mix are Samanvaya's own measurements on 24 targets [EXTERNAL]. The v9 audit varied θ from 1.0 to 2.0 and kept the capitulation edge at every value; none beat 1.5 in its worst era [EXTERNAL].

CHAPTER 6

The ladder and the four tapes

6.1 The ladder of timeframes

AFTER THIS CHAPTER YOU CAN

- list the frames each ladder reads on your chart
- explain how a lower frame is read without sampling only its last bar, and a higher one without repainting
- recognise a fallback (↺) and the intrabar depth limit

Both instruments are read across a fixed ladder of standard frames, the same on every venue. A frame is used only if the larger of the two holds a whole number of the smaller's bars; a reconstruction that split bars across bars would be inventing data.

Table 6.1 Which frames each ladder reads

Chart	Ladder down (conviction default)	Ladder up (value default)
3m	1m · 3m	4h · 1h · 30m · 15m · 3m
5m	1m · 5m	4h · 1h · 30m · 15m · 5m
15m	1m · 3m · 5m · 15m	4h · 1h · 30m · 15m
30m	1m · 3m · 5m · 15m · 30m	4h · 1h · 30m
1h	1m · 3m · 5m · 15m · 30m · 1h	4h · 1h
4h	1m … 1h · 4h	none → falls back ↺
D	1m · 3m · 5m · 15m · 30m · 1h · 4h · D	W · D
W	1h · 4h · D · W	none for conviction ↺; value M · W

Ladder down asks whether the chart's conviction is *built* from the horizons inside each bar. It is faster than chart conviction. **Ladder up** asks whether it is *held* by the horizons that contain the chart, and is slower. With the defaults (conviction down, value up) every reading spans three horizons: inside the bar, the chart, and above it.

The value ladder keeps Samanvaya's own rungs. Ladder up carries the whole composite on every higher frame. Ladder down reads only the three frames nearest below (daily reads

30m · 1h · 4h · D), breadth leg only: the RV leg cannot be re-run per rung inside Pine's request budget.

How a lower frame is read

Sampling a 1-minute rung at the daily close would describe the last few minutes of the day. Instead each lower frame runs the full engine on its own history, and the chart bar takes the **participation-weighted average** of that frame's reading across all its bars inside the chart bar: the bar seen from inside. TradingView caps the intrabars one request carries, about 100,000 on most plans, so the finest rungs reach back less far than the chart. Older bars run without them, and the panel's `rungs live` row says how many rungs a bar carried.

One disclosed drift. The depth cap counts back from today. As days pass, the oldest bars fall out of a fine rung's reach. After a reload, a ▲ ▼ at the left edge of that coverage can appear or vanish. A bar inside every rung's reach never changes.

How a higher frame is read

A higher frame read the usual way is a staircase, or it repaints. Pragati reconstructs it instead:

- from the parent frame comes only its **settled** state as of its last closed bar;
- from the chart comes the parent bar now forming, aggregated from the chart's own bars.

The result lands on the parent's settled value at its close and is identical in history and in realtime. The panel's `ladder · rebuild error` row reports how closely it landed, in points.

Fallback ↺. A direction with no frames on this chart reads the other direction instead of pausing every signal, and the tape's label shows ↺. That happens with Ladder up on 4h, 45m or monthly, and with Ladder down on 1m. Equal weight goes to every rung. The standard frames are not evenly spaced, so the tape leans toward wherever rungs are dense. That is the price of a familiar ladder, stated rather than hidden.

6.2 The four tapes and the amalgamated view

AFTER THIS CHAPTER YOU CAN

- read all four tapes' hue and intensity without a legend
- tell WIDENING from REVERTING, and RUNNING from PAUSING
- read the amalgamated view's TALL tapes

```
conviction tape (+82)   100 · tanh( mean over the ladder of raw ÷ 3σ )       chart rung incl
value tape (−82)        100 · softbound( 0.5 · mean over the ladder of z )
momentum tapes (±89)    Ladder down: tape − chart reading     Ladder up: chart reading − tap
```

Each rung is fused in linear z-space and compressed once, so agreement across frames is **rarer** than any one frame's reading. A bright tape means the whole ladder is stretched, not one horizon.

Table 6.2 The four tapes

Tape	Hue	Intensity	What it answers
MTF conviction (+82)	Green buyers, red sellers	Faint inside ±30, a clear step brighter past it, solid past ±60	Who controls across the ladder, and how far
MTF conviction momentum (+89)	Which way the faster view pulls	Near-solid while the pull grows past k·σ; moderate while it grows below; pale while it shrinks	Ladder down: BUILDING or TURNING INSIDE the bar. Ladder up: RUNNING AHEAD or a PULLBACK
MTF value (−82)	**Green rich, red cheap**	Faint inside θ, a step brighter at θ, solid toward ±70	Rich or cheap across the ladder
MTF value momentum (−89)	Green toward rich, red toward cheap	Near-solid past 0.5σ of its 200-bar history while growing	WIDENING (agrees with the tape) or REVERTING (opposes it, the fast end turning back to fair)

Phases: how the grid reads the momentum tapes

Each momentum has a standing side that changes only past its own knee: k·σ for conviction momentum, 0.5σ for value momentum. Until a knee is calibrated, the side is the sign. That side, read against its level tape's edge, is the phase the grid's WHY cell prints:

Table 6.3 Confirmed and unconfirmed phases

Instrument	Confirmed phase	Unconfirmed phase
Conviction (rows)	**Running**: momentum on the row's side. Buyers or sellers are pressing	**Pausing**: momentum against the row. A pullback inside the control
Value (columns)	**Reverting**: momentum against the dislocation. Turning up from cheap, down from rich	**Widening**: momentum with it. Still cheapening, still richening

Note the opposite rules. Conviction's edge is confirmed by **the same** hue outside it; value's by **the opposite**. Inside the capitulation cell, value reverting beat widening in every era `EXTERNAL`. That is why the ▲ waits for it.

Amalgamated

Each momentum setting can read `Separate band` (default), `Amalgamated` or `Off`. Amalgamated draws the grid's fold for you. Past its edge, a level tape stands **TALL** while its momentum confirms the edge, and short while it does not. Inside its edge it stays short and faint.

Table 6.4 Amalgamated tapes

The tape looks	It means	Grid reading
Short, faint	No edge	FAINT row, or FAIR column
Short, bright	An edge, unconfirmed	Pausing, or widening
TALL, bright	An edge, confirmed	Running, or reverting

A bright red conviction tape over a TALL red value tape is the ▲'s condition, drawn: sellers in control, price cheap, the cheapening turning up. Amalgamated drops the bands' shading

and the early TURNING INSIDE read, both unmeasured. Keep Separate if you use them. Switching a band off hides the band, never the reading.

6.3 The trace and its rails

AFTER THIS CHAPTER YOU CAN

- choose between the three traces knowing what each changes
- say what reads the trace and what always reads conviction

Table 6.5 The three traces

Trace	What the pane draws	Rails
Conviction × Value (default)	Both ingredients in σ, blended equally: 100 · softbound(0.5 · (z_c + z_v) ÷ √(2(1 + ρ))), with ρ their correlation over 200 bars, which restores the blend's variance. How **stretched** the move is, in effort and in price against fair	±θ (±43) and ±60, a 3σ stretch
Conviction	The single-frame engine alone, bit-identical to Nishchaya v3, on which the original evidence was measured	The zones, ±30 and ±60
Value	Samanvaya's value on this chart alone	±θ and ±60

The blend dims while its ingredients point opposite ways: cheap with buyers pushing is accumulation; rich with sellers pushing is distribution. It also dims while the value basket settles. A dim blend near zero is not necessarily quiet; the panel's `trace` row gives both ingredients in σ and their ρ. Calibration on Pragyam's panels: the blend sits past θ on 19–21% of days, as value alone does, and past ±60 on only 0.1–0.2% EXTERNAL.

What follows the trace, and what never does

Table 6.6 The trace's reach

Follows the drawn trace	Always reads conviction
The histogram's columns and tiers	The conviction ladder's chart rung and its momentum
The ◆'s impulse gate and pullback	The ◆'s control test (conviction above zero)
The rails and the panel's headline	R and H pivots, the quiet regime
	The grid, its push gate and the ▲▼: the Trace setting never moves a ▲ or a ▼

As a position the blended trace and its histogram carry almost no timing information once scored without look-ahead EXTERNAL. Read them as description: how far the move is stretched, and which way it is pushed.

CHAPTER 7

The grid engine and the signals

7.1 Rows, columns and the row memory

> **AFTER THIS CHAPTER YOU CAN**
>
> - say exactly when the grid's row moves, and when it is HELD
> - explain why the column has no memory and the row does

The grid reads the two level tapes into a 3 × 3 map (chapter 11). It is readable only on a standard chart with both level tapes ready.

```
target row   UP   if conviction tape ≥ +30      DOWN if ≤ −30      FAINT between
column       RICH if value tape ≥ +43 (θ)    CHEAP if ≤ −43      FAIR between
```

The column follows the value tape directly, every bar. **The row has a memory.** It moves to its target only on a push gate (section 4.3) toward it:

1. **Target above the row, and the gate is +1**
   The row moves up to the target.
2. **Target below the row, and the gate is −1**
   The row moves down to the target.
3. **Anything else**
   The row stays where it is. If the target differs, the row is **HELD**, and WHY says so in gold.
4. **No gate yet (warming)**
   The row simply equals its target, with no memory. WHY appends `gate warming`.

This is why a tape that slips briefly across ±30 does not flip a family. Pragyam's grid ignores a tape move that conviction's own histogram has not confirmed. A row move takes both the tape crossing and a push that way, on the same bar or later.

> **Held is caution, not direction.** A held row means the tape has left the row and the histogram has not confirmed a push that way. The family's units stay between the held row and half-way to the neighbouring row, according to how firmly the push holding it is drawn.

7.2 The 5 × 5 and the graded units

> **AFTER THIS CHAPTER YOU CAN**
> - compute a name's units from its position on the map
> - explain what PAUSING and WIDENING do to the units

Each of the nine families has units (chapter 11), but STATE rarely prints a cell's round number. The name is placed on the map at a **position**, and the units are read there, bilinearly between cells, as Pragyam does.

The position

1 **How intensely each tape is drawn**
A tape's ink runs from 0.12 to 0.45 inside its knee (±30 for conviction, θ for value), steps to 0.65 at the knee, and rises to 1.0 toward solid (±60 for conviction, ±70 for value). The row position is the row times conviction's ink; the column position is the column times value's ink.

2 **Half-way for an unconfirmed edge**
A row that is PAUSING, or a column that is WIDENING, is moved half-way toward the middle: the 5 × 5's in-between cells.

3 **Held rows**
A HELD row sits between itself and half-way to its neighbour, by how firmly the push is drawn.

Reading the units at a position

```
rows −1 … +1 (DOWN … UP), columns −1 … +1 (CHEAP … RICH)
units = bilinear interpolation of the nine cells at (row position, column position)
```

Worked (constructed). Sellers in control, the conviction tape at about −45 and running: row position −0.8. Price cheap, the value tape at about −62 and reverting: column position −0.9. The four surrounding cells are DOWN·cheap 4, DOWN·fair 1.5, FAINT·cheap 1.5 and FAINT·fair 1.

```
along the column   DOWN row:  0.9 × 4   + 0.1 × 1.5 = 3.75
                   FAINT row: 0.9 × 1.5 + 0.1 × 1   = 1.45
along the row      0.8 × 3.75 + 0.2 × 1.45 = 3.29 units
```

STATE prints `Buy 3.29`. If value momentum were still widening, the column position would halve to −0.45, and the same name would read 2.35 units. The tooltip shows the unshaded cell too; here, 4.00.

Measured. The units table is v8's, re-measured in v9 and kept. The graded units, read as a position, were positive in all three eras outside crypto EXTERNAL. Buy · capitulation went from 3 to 4 units in v9.1, because at 4 the grid improved or tied in all three eras on both scorers EXTERNAL.

7.3 The signals, exactly

AFTER THIS CHAPTER YOU CAN

- state the ▲ and ▼ conditions completely
- explain why a ▲ can come days after the grid entered the cell, or never
- list what pauses each family, and how the pane tells you

Table 7.1 The signal families

Signal	Fires on the first closed bar where	Also needs	Evidence
▲ Capitulation turn	the grid stands in DOWN · cheap **and** value momentum is REVERTING, having not been so on the bar before	A standard chart; both level tapes and conviction warmed up; the bar before readable; the long cooldown (10 bars) clear	EXTERNAL +0.046 to +0.056σ over 10 bars, every era, daily; positive weekly
▼ Distribution	the grid enters DOWN · rich (sellers in control of a rich price)	The same, with the short cooldown	EXTERNAL As a state, underperformance in every era; too rare to measure as an event
◆ Continuation (off)	conviction above zero; the histogram dipped below zero within 6 bars and now crosses +k·σ; conviction tape ≥ +30; value tape below +θ; effort not absorbed	The full stack, including the trace's warm-up	EXTERNAL Negative in both eras outside crypto
R regular divergence (off)	a price higher high on a conviction lower high (zone gate), the ladder also lower, value rich (> 0) at the pivot	Pivots 5 left and 5 right: the divergence is known 5 bars late	EXTERNAL About zero in every era
H hidden divergence (off)	a price lower high on a conviction higher high; conviction tape ≤ −30; value not cheap past θ	As R	EXTERNAL Negative or mixed

Entered means entered. A grid that becomes readable already inside a cell never *entered* it. Warm-up completing is not a capitulation, so a ▲ ▼ needs the bar before to be readable too. The ▲ fires on the first bar of the *turn*, not the first bar of the cell. A market can sit in DOWN · cheap, widening, for weeks with no ▲, and the ▲ comes only when the cheapening eases. It may never come.

The declaration is the last ▲ or ▼, standing until the opposite one, with no exit. Held as a position it carried no edge EXTERNAL. The state behind the ▲ did, which is why the pane background paints the state by default: green while the grid stands in DOWN · cheap, deeper once value momentum has turned, and red in DOWN · rich.

Crypto. The capitulation reading failed on crypto, which trends, so on a crypto symbol its tint and its ▲ are drawn in gold. Nothing that fires changes.

Nothing pauses silently. A tape that is warming or switched off, no frames after fallback, or Samanvaya's model still building pauses the families that read it. A gold `SIGNALS PAUSED` label at the pane's right edge names the layer. While only the trace is warming, the ▲ ▼ run and the label names just ◆, R and H.

CHAPTER 8

The auction engine and Accepted momentum

8.1 The rolling profile and migration

AFTER THIS CHAPTER YOU CAN

- say which window the trail uses with Accepted momentum on and off
- draw the five migration states from two value areas

The auction engine builds its own volume profile from the chart's bars. It does not read the session profile, whose settings and session change with the timeframe. Its window is **126 bars, six months, with Accepted momentum on (the default)**, and the conviction Lookback (20 bars) with it off. The ribbon, the probes, the AUCTION row and the trail all use that same window.

1 **Weights**
Each bar contributes its capped participation weight w (section 4.1), spread evenly from its low to its high. Raw volume let one expiry print own the POC for the whole window; flips were 2.0× likelier on the bar such a spike left the window, a flip caused by nothing.

2 **An absolute grid**
Bins are a power of two of the tick nearest ATR(20)/4, at most 300. A bin edge moves only when volatility changes by about √2, never because the window's low did. Anchored at the window's low, 29% of the value area's bin-sized moves had been phantom.

3 **POC and value area**
The POC is the heaviest bin; ties go to the bin nearest the middle. The value area starts at the POC and adds the heavier neighbour, above or below, until it holds 70%. VAL and VAH are its edges.

Migration: this window against the window N bars ago

Table 8.1 The five migration states

State	Condition (first match)	Reads
+2 · Migrating ↑	Today's VAL above the old VAH	The value areas no longer overlap. Value has moved up
+1 · Shifting ↑	Today's POC above the old VAH	The centre of value has left the old value area
0 · Balance	Anything else	Value overlaps where it was
−1 · Shifting ↓	Today's POC below the old VAL	The mirror
−2 · Migrating ↓	Today's VAH below the old VAL	The mirror

With a 126-bar window the comparison is with the value area of six months ago: a slow, structural question. In a trend, migration usually reads ±1 or ±2 for months. NIFTY 50 read Shifting or Migrating ↑ on 189 of its last 300 bars, and Balance on only 21 MEASURED.

Measured (setup lab S5, section 16.2). With Accepted momentum agreeing, bars Migrating with the trail did better over 20 bars than bars in Balance, in all three sets. The result was consistent but not confirmed MEASURED.

8.2 The acceptance trail: start, step, probe, flip

AFTER THIS CHAPTER YOU CAN

- apply the trail's four rules to a bar
- explain why the default trail can sit far from price for months
- read the trail's colour with Accepted momentum on and off

The trail is the ratcheted far edge of value: the value-area low while up, the high while down. It is the floor or ceiling of where the market has accepted price. It is updated on closed bars only and never repaints.

Table 8.2 The trail's rules, checked in this order

Rule	While up (D = +1); down mirrors	Effect
Start	Migration ≥ +1, push gate +1, close ≥ VAL	D = +1, level = VAL
Flip	Close below the level, **and** the window's POC below it, **and** migration ≤ −1	D = −1. The level is unchanged: the broken floor becomes the ceiling
Probe	Close below the level without both of the flip's other conditions	Nothing changes; the bar is a probe. Its first bar is marked with a gold ×
Step	Migration ≥ +1, push gate +1, VAL above the level, close ≥ VAL	Level = VAL. A floor is never lifted above price

So the trail moves only when value shifts its way **and** conviction confirms a push that way: the same gate by which the grid moves a row. It flips only on **acceptance**, meaning price beyond it, the market's value (the POC) following, and value having migrated that way. Price

beyond it without all three is a **probe**: in Market Profile terms, price outside value that has not been accepted.

Why the trail can sit far from price

A step needs the push gate open, and the quiet regime shuts it (section 4.2). In a long, orderly trend conviction's σ can sit in its bottom fifth for months, and the trail cannot step at all. On gold, the trail's floor stayed at 2,640 from before August 2025 to 27 May 2026 while price traded as high as 5,586. It stepped to 4,256 on 28 May, the first open gate with value migrated MEASURED. NIFTY 50's trail sat at 24,064 for the whole of its last 300 bars (chapter 25).

> **With Accepted momentum on, the trail's level is not a stop.** What trades is the side (with momentum's size), not the distance to the line. A trail 30% below price is still long. For stops, use section 21.2.

Colour

Table 8.3 The trail's colour

Accepted momentum	Hue	Intensity	Neutral or gold
On (default)	The side	The size: solid at 0.5 and above, strong from 0.25, faint below	**Neutral grey when flat**: the trail and momentum disagree
Off	The side	The conviction tape in the trail's direction: solid past ±60, strong past ±30, faint against	**Gold** when the value tape is stretched past θ its way: the move is already paid for

The value ribbon is the rolling value area as a slate band with its POC dotted, over the last 1,500 closed bars. Inside the band is **in value**; above or below is price the window has not yet accepted. Probe marks keep the latest 40. A probe lasts about four bars, and marking every bar of it turned a quarter of the chart gold.

8.3 Accepted momentum, from the inside

> **AFTER THIS CHAPTER YOU CAN**
>
> - compute S and the position from a chart's numbers
> - say what each word in the AUCTION row's CHART cell means

```
for h in 21, 63, 126, 252 bars:   s_h = clip( ln(close ÷ close h bars ago) ÷ (σ·√h), −2, +2
S        = mean(s_h) ÷ 2                      in −1 … +1   (σ: EWMA of daily returns, span 6
position = side × max(0, side × S)           side = the 126-bar trail's D
weight   = position × 10% ÷ annualised volatility, capped at 2
```

The **side** is where value was accepted over six months. The **size** is how far and how consistently price has gone over one, three, six and twelve months, each in its own volatility. Where the two definitions of trend agree, the position is the trail's side at momentum's size. Where they disagree, it is **flat**.

Table 8.4 The AUCTION row's CHART cell with Accepted momentum on

CHART cell	Meaning
`LONG 0.62`	Trail up, S = +0.62: long at size 0.62
`SHORT 0.40`	Trail down, S = −0.40
`FLAT · disagree`	The trail and S point opposite ways
—	Not ready: the trail has not started, or S needs 252 bars and σ 60

The cell is live on the forming bar and final at the close. The tested system decides at the close, trades at the next open, and rebalances weekly with a 10% band (section 18.1). Its position alert, through the one `Any alert() function call` alert, fires on each change of LONG, SHORT or FLAT. It is off by default, because an alert invites trading one chart, and the tested edge is a portfolio's.

CHAPTER 9

The session profile and open interest

9.1 The session volume profile

AFTER THIS CHAPTER YOU CAN

- say which session the profile covers on your timeframe
- read POC, VAH, VAL, HVN, LVN, naked POCs, VWAP and IB, and know which are measured

The session profile is a price-chart overlay, display only: nothing in the engine reads it. It answers **where** volume traded in the current session.

Table 9.1 The profile's session

Chart timeframe	Session
Intraday	The day
Daily	The week
Weekly	The month
Monthly and above	Twelve months

How it is built

- **From intrabars**, by actual traded price, when `Build from intrabar data` is on and the plan delivers them. Otherwise each bar's volume is spread over its range. The panel's `Profile Mode` row reports which, and the session's intrabar share.
- **Volume or time.** Volume is used when at least half of the last 500 bars print it. Otherwise the profile is time-at-price (TPO), automatically, and the session line reads TWAP instead of VWAP.
- **Bins** are ATR(20) ÷ 4 wide (`Bins per ATR`), frozen at the session's start, or a fixed number of ticks. A bar wider than 200 bins is spread over its whole range on every st-th bin, with volume conserved.

- **POC**: the first bin to reach the session's maximum. **Value area**: 70%, walked out from the POC, the larger neighbour first.
- **HVN / LVN**: a local peak at least 1.2σ above the mean of the non-empty rows, or a local trough at or below their 10th percentile. These are heuristics; the thresholds are unmeasured.

Levels drawn

Table 9.2 Session profile levels

Level	What it is	Evidence
POC, VAH, VAL	The session's most-traded price and its 70% value area	UNMEASURED
Developing trails	How POC and value developed through the session; blank on its first bar	UNMEASURED
Prior session value	Last session's pPOC, pVAH and pVAL, drawn only across the current session	UNMEASURED
Naked POC	A prior session's POC price has not traded back through: at most 5, at most 250 bars old. Since v9.9 a session's POC joins the ledger before its first bar's touch check; before, 27% of drawn naked POCs had already been traded through	UNMEASURED (PVOI's "measured" label not re-verified)
Session VWAP / TWAP	The session's volume-weighted average price; time-weighted without volume	UNMEASURED
Initial Balance	The high and low of the session's first 12 bars, intraday only: on daily bars 12 bars would swallow the week	UNMEASURED
Delta split	Each row split into inferred buying and selling, by close location per bar or the tick rule per intrabar. An inference, not order flow	UNMEASURED

The panel's `Location` row places price: above, below or in value, or at the POC (within one bin). Off value is gold, because unaccepted price can resolve by rejection or by acceptance, and the colour does not pretend to know which.

9.2 The open-interest engine

> **AFTER THIS CHAPTER YOU CAN**
>
> - explain how ΔOI becomes a σ figure and a character
> - recognise every feed state, and why an expiry is not conviction
> - say when the OI row should be believed

Volume answers *how much* traded. Open interest answers *what kind*: whether the trading created open risk or destroyed it. The sign of ΔOI is the information volume cannot carry.

1 **The companion series**
Built from the chart's prefix and ticker as `PREFIX:TICKER_OI`, then two alternates. It is never built from the tickerid, which on a modified chart is a JSON envelope. It is requested on the same timeframe, with no look-ahead.

2 **Is it live?**
At least a fifth of the last 50 bars must carry a change; otherwise the row reads STALE. An end-of-day OI series on a 5-minute chart is a step function.

3 **The build figure**
The bar's log change, divided by the RMS of the last 50 accepted changes, measured before this bar joins. RMS keeps zero at zero, so the sign is the sign of ΔOI. It needs 10 accepted changes first; until then the row reads WARMING.

4 **Structural bars**
A bar that moves more than half the book, or one beyond 3σ that also moves more than a fifth of it, is an EXPIRY / ROLLOVER. Such bars are kept out of the baseline, so one roll cannot inflate σ for the next fifty bars.

5 **The character**
Beyond ±0.5σ, the sign of ΔOI with the sign of the close's change: long build-up, short build-up, short covering, long unwinding. A book that moves while the close does not reads PRICE FLAT.

6 **Trust**
ρ(|ΔOI|, activity) over 50 bars must reach 0.20, or the row dims. Activity is volume, or true range without it. The signed correlation would be wrong by construction: OI rises on long and short builds alike.

Table 9.3 OI feed states

Feed state	Meaning	Colour
live	A reading	By character
warming	Building its baseline (10 changes)	Gold
stale	Fewer than a fifth of recent bars change	Gold
daily only	OI exists on daily bars only	Gold
none	The symbol has no open interest at all, as on cash equity and spot	**Neutral**: the correct answer
no symbol / synthetic	A malformed ticker, or a non-standard chart	Gold

Crowded: the book's level at or above the 80th percentile of its last 250 bars, which prints gold in WHY. **The gold cast** on the histogram, on by default, tints a column toward gold when exits paid for its push: covering outnumbering long build-ups above zero, or unwinding outnumbering short build-ups below, with at least a third of the window exits. It applies only with the feed live, outside structural bars, and trusted. Pushes it marks did worse than others in both eras on 240 NSE F&O stocks EXTERNAL.

The four-quadrant reading is a practitioner's framework. The academic evidence lives at a monthly horizon (Hong and Yogo 2012). Nothing validates the bar-level quadrant. "Self-limiting" is a hypothesis this table colours by, not a result UNMEASURED. On a continuous

front-month future the book falls into each expiry as positions roll, and a fall below the structural cut can read as covering or unwinding. Treat OI in expiry week with caution.

CHAPTER 10

Timing, repainting and limits

10.1 What is live, and what is final

AFTER THIS CHAPTER YOU CAN

- say for every element whether it can change on the forming bar

Table 10.1 Repaint posture, element by element

Element	On the forming bar	At the close
Trace, histogram, all four tapes	Live: they move as the bar forms	Final
Grid STATE, pane background (grid state)	Live: a tint can come and go	Final
▲ ▼ ◆, R, H, the declaration, every alert	Not drawn	Committed; never revised
Acceptance trail, probe marks, value ribbon	Not drawn for this bar	Committed; never revised
AUCTION row, Accepted momentum's CHART cell	Live; an accepted break reads "accepted, flips at the close"	Final
OI row	Live	Its baseline is committed at the close
Session profile, developing session	**The one disclosed exception**: it develops	—
Divergence markers	Not drawn; they sit on the confirming bar, 5 bars after the pivot	Committed

Higher frames enter only as their last closed bar plus chart bars that exist; lower frames only as intrabars that exist. History and realtime compute the same numbers. The rule that follows is simple: **decide at the close**.

10.2 Warm-ups, budgets and where evidence stops

AFTER THIS CHAPTER YOU CAN

- know how much history each part needs before it reads
- know which timeframes and asset classes carry evidence

Table 10.2 Warm-ups

Part	Needs	Shows while waiting
Conviction and the zones	The 200-bar Normalization window	The trace builds
Push gate, grid row memory	About 450 bars	gate warming
Value tape	Samanvaya's model: ten anchor periods past a twelve-period hold-out, and the basket settled	Grey tape; SIGNALS PAUSED names it
▲ ▼	Both level tapes and conviction's warm-up	SIGNALS PAUSED
Acceptance trail (126)	The gate, then 126 bars, then a start condition	AUCTION reads no trail yet
Accepted momentum	The trail, 252 bars for S, 60 for σ	CHART reads —
OI	10 accepted changes	warming

Budgets

Pine allows 40 unique data requests. Samanvaya spends 22; the ladders up to 10; the profile 1; OI 4. At most 37 execute on any chart, so nothing more can be requested. The plot budget is full at 64 of 64, which is why the momentum tapes' readings live in labels and the panel rather than the Data Window.

Where the evidence stops

Table 10.3 Limits

Question	Answer
Timeframe	Daily bars carry the evidence. The ▲ and the grid were also tested on weekly bars. Nothing was measured intraday: twenty years of intrabars exist on no free feed
Costs	At one ATR of stop a round trip costs about 0.02R daily, 0.045R hourly, 0.05R on 15 minutes and 0.09R on 5 minutes, against a best measured edge near 0.05R. Daily is the only timeframe with real headroom EXTERNAL
Asset classes	The ▲ and the grid were positive in 4 to 5 of the 5 non-crypto classes per era; crypto failed EXTERNAL. Accepted momentum was tested on 119 markets in six classes, then 411 NSE stocks and crypto MEASURED
Tuning	900 randomised parameter sets: fitted edge and unseen edge correlated at about zero. Lookback ≥ 20 and the inner zone ≤ 40 were the only preferences that held. **Do not tune Pragati to a backtest** EXTERNAL
Charts	Heikin Ashi, Renko, Kagi, Point & Figure and range bars hand every engine prices that never traded. Everything pauses and says so

PART III

Reading every state

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CHAPTER 11

The Conviction-Value Grid

Rows come from the conviction tape: UP past the inner zone (+30), DOWN past −30, FAINT between. Columns come from the value tape: CHEAP past −θ, RICH past +θ, FAIR between. Each of the nine cells is a family named for what it says to do. The number is its units, the relative weight a portfolio gives the name.

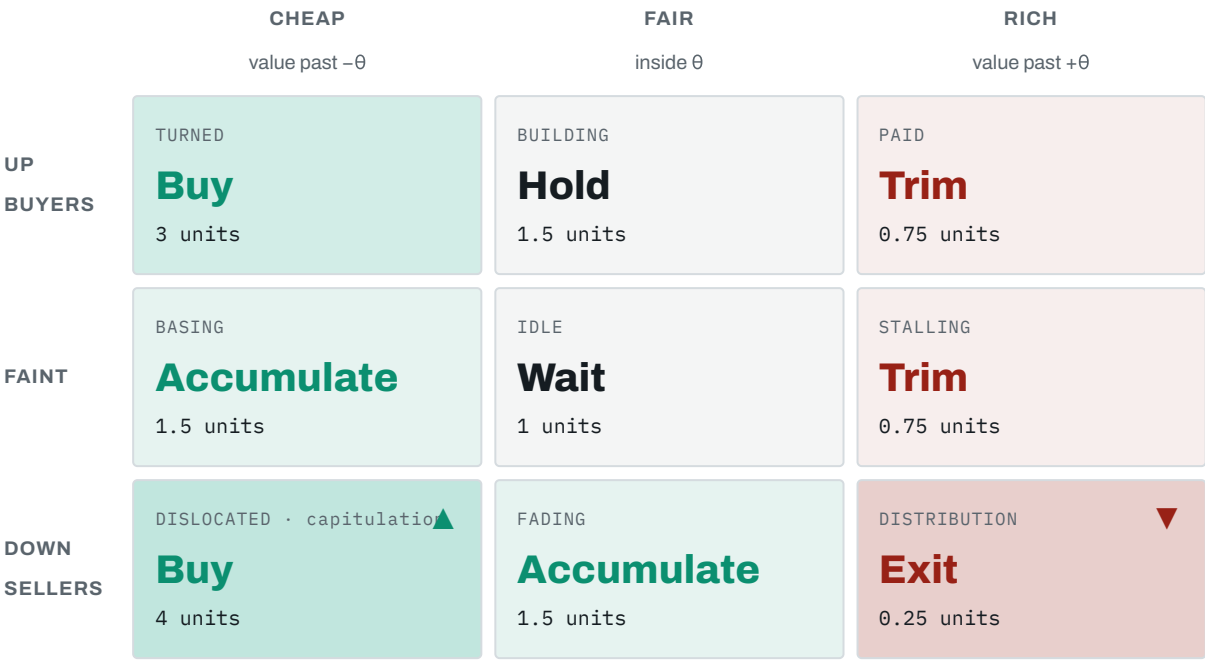


Figure 11.1 The 3 × 3 map. Green actions buy, red actions sell, grey change nothing. The word is the weight: Buy 4 is four times Wait 1. EXTERNAL The graded units were positive in all three eras outside crypto.

Rows move only on a push

A row changes only when conviction's own histogram confirms a push that way: on the move's side, not turning, not in a quiet regime. Without that push the row is **held** where it was, and WHY shows "held" in gold. This stops the grid flickering on every wobble of the tape.

The 5 × 5: phases

Each edge has a momentum that confirms it or not. Conviction confirms a row while buyers (or sellers) are **running**; it is **pausing** otherwise. Value confirms a cheap or rich column once it is **turning** back toward fair; it is **still cheapening** or **still richening** otherwise. An unconfirmed edge sits half-way to the middle, so its units fall. Inside a cell the units are also shaded by how strongly each tape is drawn.

Table 11.1 Phases in the WHY cell

You see in WHY	It means	Effect on units
buyers running · sellers running	Conviction momentum stands on the row's side	The full row
buyers pausing · sellers pausing	A pullback against the row	Half-way to the middle
turning up (from cheap) · turning down (from rich)	Value momentum reverting toward fair	The full column. On DOWN · cheap this is the ▲'s condition
still cheapening · still richening	The dislocation is still widening	Half-way to the middle
no edge	FAINT and FAIR: nothing to confirm	Wait, 1 unit
· held (gold)	The tape left the row without a push that way	Between half and all of the held row, by the push's strength
· gate warming	Conviction's histogram is still calibrating (about 450 bars)	Rows follow the tape with no memory yet

CHAPTER 12

The readout table

Five columns ask the same question on every row. STATE: where the layer stands. WHY: what makes it so. PUSH: what pushes it now. CHART: one more reading that differs by row. SPAN: how long it has stood, counting the current bar as the first. Hover any cell on TradingView for its tooltip.

OI row (futures with open interest)

Table 12.1 OI row states

STATE	Colour	What happened this bar
`Long build-up`	Green	Price up, open interest up: new longs, risk committed
`Short build-up`	Red	Price down, open interest up: new shorts, risk committed
`Short covering`	Gold	Price up, open interest down: a rally paid for by exits. Short-covering rallies lagged `EXTERNAL`
`Long unwinding`	Gold	Price down, open interest down: a decline paid for by exits
`Balanced`	Grey	The change is within 0.5σ of zero
`Price flat`	Grey	The book moved past 0.5σ but the close did not, so there is no side
`Expiry / roll`	Gold	Over 50% of the book in one bar, or over 3σ and 20%: structural, ignored
`Warming` `Stale feed` `Daily only` `No symbol` `Paused`	Gold	Not enough history yet, an end-of-day feed, no OI at this timeframe, a chart modifier, or a non-standard chart
`No OI`	Grey	The instrument has no open interest. Correct for cash equity and spot

WHY adds who is doing it and how full the book is: "book 85 pct" means total open interest sits at the 85th percentile of its last 250 bars. From the 80th it turns gold: a crowded book. PUSH is the change in σ. A faded row means the size of the book's change does not track activity (ρ below 0.20), so do not read it. **Expiry caution:** a front-month future's book falls into every expiry as positions roll, which can read as covering or unwinding. `UNMEASURED`

CVG row

Table 12.2 CVG row cells

Cell	Shows	Example
STATE	The family's action and its shaded units	`Buy 3.42` `Exit 0.31`
WHY	Each momentum's phase on its edge (table above)	sellers running · turning up
PUSH	Conviction's histogram: ↑ ↑ an impulse past k·σ, ↑ a push, turning (no push), quiet (gold, no push), and the mirrored ↓ ↓↓	↑
CHART	Where this chart alone would put the name, against the ladder's state. ↑ more units, ↓ fewer, = same units elsewhere, same	↑ Buy
SPAN	How long the name has stood in this family	6d

AUCTION row

Table 12.3 AUCTION row cells

Cell	Shows	States
STATE	Value's migration against the window six months earlier	`Migrating ↑` value areas no longer overlap · `Shifting ↑` only the POC has left · `Balance` · and the mirrored ↓ states
WHY	The trail's side and the close against value	trail ↑ · in value · `probe, not accepted` · accepted, flips at the close · push gate warming
PUSH	The histogram push the trail steps by	↑ · turning
CHART	Accepted momentum's position and size	`LONG 0.62` `FLAT · disagree` `SHORT 0.40`
SPAN	Bars since the trail started or flipped (the position can be younger)	41d

CHAPTER 13

Panel, marks and troubleshooting

13.1 The information panel

Off by default (setting 8). Turn it on for a diagnostic view at the right of the pane.

Table 13.1 Information panel rows

Row	Read it for
Header	The trace's zone in words: buyers in control, STRETCHED UP, CHEAP · past θ, WARMING UP
declaration	The last ▲ or ▼ and how many bars ago. As a held position it carried no edge `EXTERNAL`. Paused with the reason, or off
value · chart	Samanvaya's value on this chart against θ. Gold while the macro basket settles
MTF · conv · value	The two tape readings. ↺ means the ladder fell back to the other direction
MTF momentum	Both momentum readings with their phase words
impulse	The histogram against k·σ, in native units
trace	The trace's ingredients: C and V in σ, ρ, split or settling
raw share · σ	The untransformed reading. QUIET (gold): small imbalances are being magnified
participation	What weighted the last closed bar, and by how much. At the cap: extreme participation
effort → result	How much effort became movement. ABSORBED (gold) in the bottom fifth
macro share, overlap, rungs, driver timing, rebuild error	Model diagnostics. Gold flags mean read the value side with care
Profile Mode, Location	What the session profile actually did, and where price sits in it

13.2 Signals and marks

Table 13.2 Signals and marks

Mark	Fires when	Default	Evidence
▲ Capitulation turn	The first closed bar in DOWN · cheap with value momentum reverting. Not on a grid that wakes up already inside the cell	On	EXTERNAL Positive in all three eras, daily and weekly, in 4 to 5 of 5 non-crypto classes. About +0.05σ over 10 to 20 bars: a lean, not a trade. Fails on crypto
▼ Distribution	The first closed bar on which sellers control a price rich past θ	On	EXTERNAL As a state, followed by under-performance in every era
◆ Continuation	A pullback that reset value, then a push past +k·σ	Off	EXTERNAL Negative in both eras. Do not lean on it
R · H divergence	Price and conviction disagree at pivots	Off	EXTERNAL R about zero; H negative or mixed
× Probe	The first close beyond the trail the market has not accepted	On	UNMEASURED
Background tint	While the grid holds capitulation (green) or distribution (red)	Grid state	EXTERNAL The state carried the edge
Gold cast	The push was paid for by exits (open interest)	On	EXTERNAL Marked pushes did worse

> Every signal fires at the close of its bar and never repaints. Intrabar, the readout is live and can change until the close.

13.3 When something looks wrong

Table 13.3 Symptoms, causes and fixes

You see	Why	What to do
push gate warming · no trail yet	Conviction's histogram needs about 450 bars	Load more history, or wait
non-standard chart · Paused	Heikin Ashi, Renko and similar draw prices that never traded	Switch to standard candles
OI: No OI	Cash equity and spot have no open interest	Expected. Open the futures symbol to read OI
OI: Stale feed · Daily only	An end-of-day OI feed on an intraday chart	Use the daily chart
OI: Warming	Fewer than 10 accepted changes in the feed so far	Wait a few bars
OI row faded	The book's changes do not track activity (ρ below 0.20)	Do not read it on this symbol
Grey trail · FLAT · disagree	The trail's side and momentum disagree	No position. Not an error
Gold ▲ and gold tint	A crypto symbol: the capitulation reading failed there	Skip the capitulation plan

You see	Why	What to do
held (gold) in WHY	The tape moved without a push that way	Wait for the push, or for the tape to return
↺ on a tape label	The chosen ladder direction has no frames on this chart	Expected on 1m or weekly charts
Profile Mode: TPO · auto	Fewer than half the bars carry volume	Expected on index spot
Profile Mode: bar, or a share like 5 40%	Intrabar history is limited by your plan	Normal for older sessions
The readout covers the chart	Its corner clashes with your layout	Change Readout position (setting 6)

PART IV

Reading in context

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CHAPTER 14

The context stack

14.1 The six layers, top down

AFTER THIS CHAPTER YOU CAN

- read any chart in a fixed order, so no layer is skipped and none is read twice
- say which layer can veto which

Every reading on Pragati belongs to one of six layers. Read them in this order. A higher layer sets the meaning of everything below it, and a layer that cannot be read stops the reading there.

1 · Readable?	Standard chart, warmed up, nothing paused
2 · Regime	Trend, balance, transition, quiet, dislocation
3 · Location	In value, out of value, against the trail
4 · Control and value	The grid family and its phases
5 · Flow	Open interest, crowding, the gold cast
6 · Trigger	▲ ▼, a flip, an AM change

Table 14.1 The context stack

Layer	Where to look	Questions	It can veto
1 · Readable	SIGNALS PAUSED label, grey tapes, gate warming, non-standard chart, the panel's rungs live	Is every layer I am about to read actually computed?	Everything. A paused layer is not a neutral reading; it is no reading
2 · Regime	AUCTION STATE (migration), the trail's side, Accepted momentum's CHART cell, the panel's raw-share row (QUIET)	Is value moving, and do the two definitions of trend agree?	Trade entries: no trend trade in a FLAT · disagree regime
3 · Location	The value ribbon, the trail, the session profile's Location row	Is price accepted here, or probing beyond?	Chasing: a probe is a caution on the trend (section 16.2)

Layer	Where to look	Questions	It can veto
4 · Control and value	CVG STATE, WHY and PUSH; the pane's tapes and background	Who controls the ladder, is price cheap or rich, and are both edges confirmed?	Adding: no adds into a held row or a widening dislocation
5 · Flow	OI STATE and WHY; gold-cast columns	Is risk being opened or closed, and is the book crowded?	Size, never direction
6 · Trigger	▲ ▼ on the pane; the AUCTION row's CHART changes; alerts	Has a measured event happened at this close?	—

Why this order. Readability first, because a paused layer looks like a neutral one. Regime before location, because "below value" means a probe in a trend and a rotation in balance. Location before the grid, because the grid knows nothing about where value was accepted. Flow is a size modifier and never sets direction: the four-quadrant reading is unmeasured at the bar level `UNMEASURED`. The trigger comes last, because a trigger read before its context is a coin flip with a story attached.

14.2 The five-minute top-down routine

AFTER THIS CHAPTER YOU CAN

- run one routine on every chart, at every close you act on
- write the reading down in one line

1 **Weekly chart, one minute**
Grid family and phases (the ▲ and the grid were tested weekly). Trail side. Is the weekly value tape solid either way? This is the tide.

2 **Daily chart: readability**
Any gold label, grey tape or `gate warming`? Note it, and read only what is computed.

3 **Daily chart: regime**
AUCTION STATE and the CHART cell. Write *trend up*, *trend down*, *balance*, *transition (FLAT · disagree)* or *quiet*.

4 **Daily chart: location**
Close in value, above or below the ribbon? Any probe × in the last five bars? How far is the trail, in ATRs?

5 **Daily chart: control and value**
CVG STATE with units; WHY phases; PUSH. Is anything HELD?

6 **Daily chart: flow**
Futures only: OI character and crowding. Stale, warming or untrusted means skip this layer.

7 **Daily chart: trigger**
A ▲ or ▼ at this close? Did the CHART cell change? Did the trail flip?

8 **Write one line**
For example: `RELIANCE · D · readable · trend down (SHORT 0.50) · below value, no probe · Accumulate 1.5, sellers running, still cheapening · no OI · no trigger.`

The line takes ten seconds once the routine is habit, and it is what your journal stores (section 21.4). Two weeks of lines on the same market teach more than any chapter here.

14.3 Timeframes without double counting

AFTER THIS CHAPTER YOU CAN

- choose a decision timeframe and an execution timeframe
- avoid reading the same horizon twice

Pragati already reads several timeframes inside one chart: conviction from inside the bar, value from above it (section 6.1). Opening a second chart one step up does not add a fresh opinion. A daily chart's value tape *is* the weekly and the daily. Use other timeframes for different jobs, not for more votes.

Table 14.2 Timeframe jobs

Job	Timeframe	Why	Evidence
Investing decisions	Weekly, then daily	The ▲ and the grid were measured on both; weekly is calmer	EXTERNAL
Accepted momentum	Daily, decided at the close	The only timeframe it was tested on	MEASURED
Execution	Intraday (15m to 1h)	To choose an entry inside the next session: never to change the daily decision	UNMEASURED
Swing context	Daily trail and ribbon	Value over six months	UNMEASURED
Intraday trading	5m to 1h	Nothing was measured, and costs eat most measured edges below daily	Costs: EXTERNAL

When weekly and daily disagree. The weekly grid in DOWN · cheap while the daily reads UP · cheap means a turn is starting inside a weekly dislocation. That is the picture the capitulation plan is built for, and the daily leads. A weekly Exit · distribution with a daily Buy is a bounce inside distribution: the daily does not overrule the weekly for a holding.

CHAPTER 15

Regimes, and how meaning changes

15.1 Five regimes and how to tell them apart

AFTER THIS CHAPTER YOU CAN

- name the regime from the readout in under a minute
- know what each regime does to every other reading

Table 15.1 The five regimes

Regime	How it reads	What it changes
Trend	Trail side and S agree: `LONG 0.45` or `SHORT 0.38`. AUCTION STATE Shifting or Migrating its way, for weeks	Breakouts from value in its direction did better than pullbacks into it (section 16.4). Probes against it are cautions on the trend, not reversals
Balance	AUCTION STATE Balance; S small; price rotating inside the ribbon; the trail flat or recently flipped	Edges of value are references, but no setup is measured here `UNMEASURED`. The grid's FAINT families dominate
Transition	`FLAT · disagree` : the trail and momentum point opposite ways. Often follows a flip, or a long probe cluster	Accepted momentum is flat by design: no trend trade. On the 126-bar trail, a probe with migration already toward it flipped 15–28% of the time within 20 bars, against about 4% otherwise `MEASURED`
Quiet	Panel raw-share row QUIET; desaturated columns; the push gate shut; the trail frozen far from price	Readings are amplified: a small imbalance reads large. The grid's rows hold. Do not read the trail's distance as risk `MEASURED`
Dislocation	The value tape solid past θ; the grid in a CHEAP or RICH family; pane tinted green (capitulation) or red (distribution)	The investor's regime. Capitulation followed by gains in every era outside crypto; distribution by underperformance `EXTERNAL`

Regimes overlap. A market can be in a trend and quiet at once, which is the common case for an orderly index, or in a dislocation inside a downtrend. Write both. The regime that changes your *action* is the one to lead with.

15.2 Same reading, different meaning

AFTER THIS CHAPTER YOU CAN

- translate a reading through its context before acting on it

Table 15.2 Context translates the reading

Reading	In a trend its way	Against a trend	In balance	In a quiet regime
Probe × (126-bar trail)	Caution on the trend: it did worse than usual for 10–20 bars MEASURED	A probe toward the counter-trend: still rarely accepted (under 5% in 20 bars) MEASURED	Rotation inside value	As in a trend; the trail cannot step anyway
Close out of value (beyond the ribbon), its way	The measured better entry, ahead of pullbacks MEASURED	A probe against the trend	A test of the edge UNMEASURED	Possibly amplified; size smaller
Close back into value from outside	A pullback: no measured edge over the trend's average bar MEASURED	—	Rotation	—
CVG Buy 3+ (TURNED, DISLOCATED)	A dip bought inside a trend: what the capitulation plan is built for UNMEASURED	A turn against the trend: build in stages UNMEASURED	Accumulate	Rows held; units unreliable
CVG Exit (DISTRIBUTION)	Early distribution while still long EXTERNAL	Aligned to the downside	Exit or avoid	Rows held
OI long build-up	Committed buyers UNMEASURED	Shorts are being met by new longs UNMEASURED	Inside value: positioning	—
OI short covering (gold)	A rally on exits: fragile; gold-cast pushes lagged EXTERNAL	A squeeze into a downtrend	—	—
A ▲	A capitulation dip inside an uptrend: add EXTERNAL	A capitulation inside a downtrend: the classic horizon conflict; stage small EXTERNAL	Stage in EXTERNAL	Fires only on a readable grid; trust it as usual

The table's left column never changes on the chart. Its meaning does. When two columns seem to apply, use the stricter one.

15.3 Asset classes

AFTER THIS CHAPTER YOU CAN

- know what changes from one class to another before trusting a reading

Table 15.3 Class notes

Class	What to know
Indices	Spot indices have no volume: participation falls back to relative true range, and the profile to TPO. Accepted momentum was tested on 16 Indian indices. Index futures carry OI
NSE stocks	The bulk of the Accepted momentum holdout (391 stocks) MEASURED and of the OI evidence (NSE F&O) EXTERNAL. Today's listings only, so survivorship applies. Costs assumed: 15 bps one way
Commodities	The ▲ and the grid were positive in commodities EXTERNAL. Continuous futures roll: expect structural OI bars at each roll
FX	INR pairs are managed and quiet: σ is low, so the tested weight (10% ÷ volatility) can reach its cap of 2. USD/INR's annualised volatility was 8.6% at the end of October 2026, so a size-0.10 position was already 0.12 of capital MEASURED. The INR pairs lost money in the Accepted momentum test MEASURED
Crypto	Capitulation **failed** on crypto EXTERNAL; its tint and ▲ are gold. Crypto trends; Accepted momentum was about flat on crypto MEASURED. Bars run seven days a week, so 300 bars are about ten months
ETFs	Read as their underlying. Thin ETFs print stale prices; check the panel's participation and the ribbon's width

PART V

Fakeouts and traps

Measured

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CHAPTER 16

Probes and acceptance

16.1 The life of a break

AFTER THIS CHAPTER YOU CAN

- follow a break from its first bar to its resolution
- say how often each outcome happens, on both trails

A **break** is a close beyond the trail. It is resolved in one of two ways. Either the market's value follows and has migrated, and the break is **accepted**, so the trail flips at that close. Or it does not, and the bar is a **probe**: a gold × marks its first bar. A probe then either flips the trail later or ends with price back on the trail's side.

Probe onset ×	A close beyond the trail; POC or migration not with it
Probe episode	A median of 4 bars on the 126-bar trail, 3 on the 20-bar
Rejected	Back on the trail's side, no flip
or accepted	POC beyond and value migrated: the trail flips

Table 16.1 What breaks become (DEV · HB · HA, setup lab)

	126-bar trail (default)	**20-bar trail (AM off)**
Probe onsets measured	6,012 · 3,019 · 31,961	11,955 · 5,706 · 62,616
Back on the trail's side within 5 bars	57% · 58% · 57%	56% · 56% · 55%
Flipped within 10 bars	2.3% · 1.9% · 1.9%	27% · 28% · 26%
Flipped within 20 bars	**4.5% · 3.8% · 4.2%**	**59% · 59% · 59%**
Median probe length	4 bars	3 bars

The two trails behave like different instruments. A six-month floor is almost never broken by a probe: fewer than one probe in twenty is followed by a flip within a month. A twenty-bar trail flips after most probes, and at about the rate the same rules produce on driftless ran-

dom walks (58%). On the short trail, a probe says little beyond the trail's own mechanics MEASURED.

16.2 What a probe does to the trend

AFTER THIS CHAPTER YOU CAN

- say what the gold × is worth, measured
- stop treating probes as fakeouts to fade

The prediction, written before the test, was that probes are fakeouts: after one, the trail's side should do at least as well as it usually does. It came out the other way, in every set and on both trails MEASURED.

Table 16.2 After a probe onset (setup lab F1)

Outcome on the trail's side, beyond its average bar	DEV (2006–19)	HB (2020–26)	HA (411 unseen markets)
126-bar trail, next 10 bars	−0.076σ (t −3.0)	−0.048σ (t −1.9)	−0.070σ (t −8.5)
126-bar trail, next 20 bars	−0.092σ (t −2.2)	−0.038σ (t −1.4)	−0.080σ (t −8.7)
20-bar trail, next 10 bars	−0.056σ (t −3.4)	−0.039σ (t −2.3)	−0.048σ (t −8.2)

The probe is a caution on the trend, not a fakeout. On the default trail it almost never flips the side, but the side's next two to four weeks are weaker than usual, by about 0.05 to 0.09σ. That is exactly what gold means in Pragati: *smaller, or later*. Do not add on a probe. Do not fade it either (next section). Hold, and let the trail decide.

The sizes are small, as every honest edge here is: a lean measured across thousands of events, worth acting on only as a size adjustment across many markets.

16.3 What gets accepted: the fakeout checklist

AFTER THIS CHAPTER YOU CAN

- score a break against four conditions at the close
- know which conditions matter on which trail

Four features of the break bar were tested, each predicted in advance to make acceptance likelier. A break that will be accepted is the opposite of a fakeout.

Table 16.3 The four conditions (setup lab F2; DEV, HB and HA all reported in docs/setup-lab.md)

At the probe's first bar	20-bar trail: flips within 20 bars	Verdict	126-bar trail	Verdict
Depth: the close at least 0.5 ATR beyond the trail	71% vs 54% (HA; t 37.5)	**Confirmed** MEASURED	4.3% vs 4.1%	Not supported
Weight: participation w ≥ 1.5	62% vs 57% (HA; t 9.7)	**Confirmed** MEASURED	3.2% vs 4.5% (fewer)	Not supported (opposite)
Push: the push gate against the trail	60% vs 55% (HA; t 8.6)	Consistent MEASURED	3.4% vs 4.8% (fewer)	Not supported (opposite)
Migration already toward the break	62% vs 58% (HA; t 5.1)	Consistent MEASURED	15–28% vs about 4%	**Consistent** MEASURED

Table 16.4 Flip rate within 20 bars by the number of conditions met

Conditions met	0	1	2	3	4
20-bar trail, HA (62,616 probes)	51%	56%	62%	71%	72%
20-bar trail, DEV	52%	57%	64%	73%	77%
20-bar trail, HB	49%	57%	63%	69%	66%

The checklist

- [ ] Which trail? On the **126-bar** default, only migration matters: is AUCTION STATE already Shifting or Migrating toward the break? If not, the break is very unlikely to be accepted within a month.
- [ ] On the **20-bar** trail, count: depth ≥ 0.5 ATR · participation ≥ 1.5 · push against the trail · migration toward the break.
- [ ] Zero or one condition: about one break in two is accepted, the rate of chance. Two or more: about two in three. Three or more: about seven in ten.
- [ ] Whatever the count: a probe on the 126-bar trail is a caution on the side you hold (section 16.2).

On the six-month trail, hard and heavy breaks were *less* likely to flip it. A six-month floor is not broken by one day's force; it is broken when value moves.

16.4 Entries around value, measured

> **AFTER THIS CHAPTER YOU CAN**
> - say which of four entry rules carried timing beyond the trend itself

Four entries were tested on the 126-bar trail. Each was measured against the trend's own average bar, so a positive number is timing *beyond* simply being on the right side.

Table 16.5 Four entries (σ beyond the trend's average bar; setup lab S1–S4)

Entry at the close	10 bars: DEV · HB · HA	20 bars: DEV · HB · HA	Verdict
Rejection: the first bar back on the trail's side after a probe	−0.03 · +0.02 · −0.04 (t −5.3)	−0.06 · +0.05 · −0.04	Not supported MEASURED
Acceptance: the flip bar, in the new direction	+0.02 · +0.09 · +0.08 (t 4.4)	+0.05 · +0.11 · +0.11 (t 6.7)	Consistent MEASURED
Pullback into value, AM agreeing	−0.00 · +0.01 · −0.00	−0.01 · −0.05 · −0.01	Not supported MEASURED
Breakout from value, AM agreeing	+0.06 (t 4.1) · +0.05 (t 2.5) · +0.06 (t 10.4)	+0.04 · +0.04 · +0.05	Positive everywhere; not predicted MEASURED

> **With the trend, the break of value beats the return to it.** A close out of value in the trend's direction, with Accepted momentum agreeing, did better than the trend's average bar in all three sets. A pullback into value did not, and the difference between the two was significant everywhere (t −2.5 to −11.5). Buying the dip into value is the intuitive trade, and it measured nothing here. Breakouts were not predicted, so treat the result as strong evidence that a fresh test must still confirm.

Hit rates sit between 50% and 54%. None of these is a setup that wins most of the time. They change *when* you act inside a trend that Accepted momentum already holds, and by how much.

CHAPTER 17

The trap catalogue

17.1 The trap catalogue

AFTER THIS CHAPTER YOU CAN

- recognise twenty-one ways to lose money with this indicator, and the evidence behind each

Table 17.1 Twenty-one traps

Trap	Why it tempts	What the evidence says	Instead
Chasing the probe ×	A close beyond the trail looks like a breakout	On the 126-bar trail fewer than 1 in 20 flip it within 20 bars MEASURED	Wait for the flip; treat the probe as a caution on your side
Fading the probe	Price back on the trail's side looks like a failed break to buy	The rejection entry measured nothing, negative in HA MEASURED	Hold what the system holds; do not add on the rejection
Buying the dip into value	"Value is support"	Pullbacks into value: no edge; breakouts from value did better MEASURED	Prefer the close out of value in the trend's direction
The forming bar	A ▲, a tint or a probe appears intraday	Live readings can vanish before the close (section 10.1)	Decide at the close
The trace as a timing signal	The trace turning up looks like a buy	The trace and histogram carry almost no timing information EXTERNAL	Read them as description
Trace and signal crossovers	A crossing looks like a trigger	k = 0 was the worst setting out of sample EXTERNAL	Ignore plain crosses
Divergence	Price higher, conviction lower	R about zero, H negative or mixed EXTERNAL	Leave R and H off
The ◆	A pullback resumed with control	Negative in both eras outside crypto EXTERNAL	Leave it off
The 20-bar trail as a trade	It flips often and looks responsive	It failed every mark of a pre-registered test MEASURED	Use Accepted momentum, or the trail as structure

Trap	Why it tempts	What the evidence says	Instead
Quiet loudness	A bright reading in a dead market	Scaling divides by a collapsed σ UNMEASURED	Check the panel's raw-share row for QUIET
Gold read as a direction	Gold looks like a warning to sell	Gold means caution only	Smaller, or later. Never short because of gold
Held rows	The family name looks current	The row is held against its tape; the histogram has not confirmed	No adds into a held row
Capitulation on crypto	A ▲ on BTC	The capitulation reading failed on crypto EXTERNAL	Do not use the ▲ or the grid's Buy 4 on crypto
Short-covering rallies	A green push	Pushes paid for by exits lagged EXTERNAL	Size down under a gold cast
Expiry OI	A collapsing book reads as unwinding	Rolls and expiries move the book mechanically	Ignore OI character in expiry week
Non-standard candles	Heikin Ashi looks smoother	Every price on it is synthetic	Use standard candles; Pragati pauses otherwise
The trail's level as a stop	The line looks like support	The default trail can freeze far from price in a quiet regime MEASURED	Stops from section 21.2, not the trail's distance
One-market Accepted momentum	It worked on the backtest of my favourite stock	On one market the median Sharpe was +0.23 to +0.38, and a quarter lost MEASURED	Run it across many markets, sized alike
Tuning to a backtest	Better settings for this stock	Fitted and unseen edge correlated at about zero over 900 sets EXTERNAL	Keep the defaults
Intraday costs	More signals on 5-minute bars	Costs reach 0.09R against a best edge near 0.05R EXTERNAL	Daily for decisions
Left-edge ▲s	A ▲ appears on old history after a reload	The intrabar depth cap slides with time (section 6.1)	Trust ▲s inside every rung's reach

17.2 The fakeout protocol

AFTER THIS CHAPTER YOU CAN

- run a five-question check before acting on any break

1 **Is this bar closed?**
If not, nothing has happened yet.

2 **Which trail, and which event?**
A flip is acceptance. A gold × is a probe. A close out of the ribbon with the trail is a break of value, not of the trail.

3 **If it is a probe against my position**
Do not exit on it, and do not add. On the 126-bar trail it is a caution: if you trade discretionarily, size down by a step, or wait (section 16.2).

4 **If it is a probe in my favour (against the trail)**
Do not enter on it. Score it (section 16.3). On the 126-bar trail, only migration toward it makes acceptance plausible.

5 **If it is a break of value with the trend**
The measured better entry. Is Accepted momentum agreeing (LONG or SHORT, not FLAT)? Then it is the moment to act if you were waiting to (section 20.3).

PART VI

Trade guide

Systematic and swing

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CHAPTER 18

Accepted momentum

18.1 The tested system

This is the one trade system in Pragati that passed a held-out test MEASURED. It was chosen by a fixed rule from eleven trials on data up to 2019, frozen, and then run once on 2020-26 and on 411 markets it had never seen.

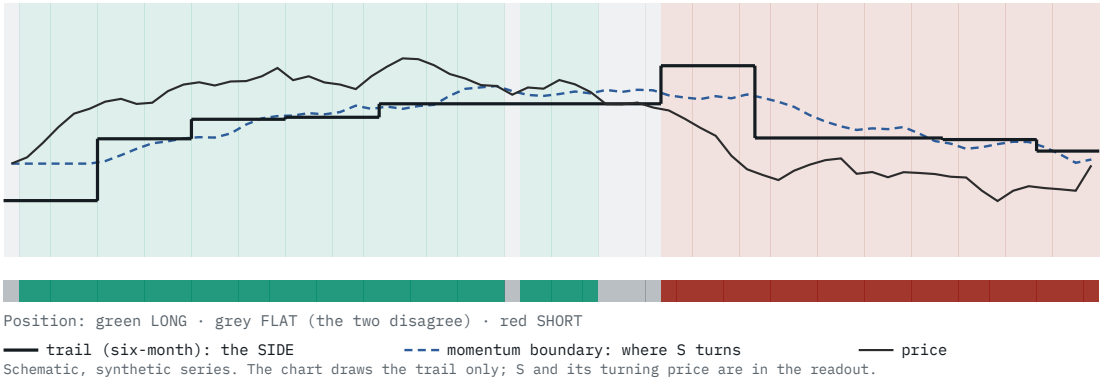


Figure 18.1 How the position is decided. The trail's side and momentum's direction must agree.

Table 18.1 Accepted momentum, the tested rules

Rule	Exactly
Side	The acceptance trail on a 126-bar (six-month) window: up while value has migrated and ratcheted up, down while it has migrated down
Momentum score S	The 1-, 3-, 6- and 12-month moves of the close (21, 63, 126, 252 bars), each divided by volatility × √horizon, capped at ±2, averaged and halved, so S lies between −1 and +1
Position	Side × max(0, side × S): LONG with size S when both say up, SHORT with size |S| when both say down, FLAT when they disagree
Decide and trade	Read at the close; trade at the next open. Never on the forming bar
Size	Weight = size × 10% ÷ annualised volatility, capped at 2 (two times the market's capital slice)

Rule	Exactly
Volatility	The test used an exponential average (span 60) of daily log returns, × √252. On TradingView the built-in Historical Volatility indicator with length 60 is a close stand-in UNMEASURED
Rebalance	Once a week (every fifth bar in the test). Each market moves to its target only if the target has moved by more than a tenth of itself. That includes going flat and changing sides: between rebalances the position is left alone

When an alert arrives. The position alert fires on the close where the side changes. Under the tested rules you act at the next weekly rebalance. Acting at the next open instead is faster, but that variant was not tested.

Worked sizing example

```
Book ₹10,00,000 across 20 markets  →  slice per market ₹50,000
Stock X: readout says LONG 0.62; annualised volatility 28%
weight   = 0.62 × 10% ÷ 28%                  = 0.221
notional = 0.221 × ₹50,000                    = ₹11,071
price ₹2,500  →  buy 4 shares (₹10,000), the nearest whole lot
Cap: weight never above 2.0 (notional ₹1,00,000 for this slice)
```

18.2 Running it as a portfolio

Table 18.2 Results on held-out data

Holdout	Sharpe, net of costs	Timing beyond leaning long	Worst drawdown	Markets positive
411 unseen markets	+0.79 (t 4.0)	t 4.1	−3.7%	79%
119 markets, 2020-26	+1.21 (t 3.7)	t 2.2	−2.5%	80%

- **The edge is a portfolio's.** On one market the median Sharpe was only +0.23 to +0.38, and about a quarter of markets lost money over the period. Run the same rules on many markets at once.
- **It did not beat plain momentum** by the pre-registered test. It matched it at lower turnover and drawdown.
- **Costs.** At twice the assumed costs the Sharpe was still +0.69 and +1.11.
- **Risk level, and real drawdowns.** At 10% per market the diversified book ran at only about 1.3% to 1.6% volatility, which is why its drawdowns above look small. Scaled to 10% volatility for the whole book, the same paths would have drawn down roughly **16% to 29%**. Expect drawdowns of that size at a useful risk level. Getting there needs futures-style leverage, and shorting Indian cash stocks overnight needs stock or index futures.
- **How many markets.** The tested books held 119 and 411 markets. With fewer, results drift toward the single-market figures. About 20 to 30 markets across several classes is a practical floor UNMEASURED.
- **Where it was weak.** The INR currency pairs lost; crypto was about flat (section 27.4).

The costs the test charged

Table 18.3 Costs charged in the test

Class	Index futures and ETFs	FX	Commodities	US stocks	NSE stocks	Crypto
One way, per change in weight	5 bps	3 bps	8 bps	10 bps	15 bps	20 bps

> **Before capital:** six months of paper trading at the same costs, then one-tenth of the intended size for six months. Stop if the drawdown exceeds 1.5 times the backtest's worst, or if the rolling 12-month Sharpe stays below zero for two quarters.

CHAPTER 19

Initiating and positioning

19.1 From the close to the order

AFTER THIS CHAPTER YOU CAN

- run the tested system's daily timeline without improvising
- handle a gap, a missed open, a holiday and a partial fill

The tested system decides at the close of bar t and holds from the open of t+1 to the open of t+2. Every result in chapter 18 was computed that way. The timeline below keeps you inside it.

Table 19.1 The tested timeline

When	What	Notes
At the close	Read the AUCTION row's CHART cell on every market in the book	A forming bar is not a decision (section 10.1)
After the close	Compute each target weight (section 19.2). Is today a rebalance day?	The test rebalanced weekly, every fifth bar; pick a fixed weekday and keep it
After the close	On a rebalance day, change only the markets whose target moved more than 10% of itself. That includes flips and going flat	Between rebalance days the position is left alone, flips included
Next open	Place the orders at or near the open	The test traded the open. A market-on-open order is the closest match

When reality does not match the test

Table 19.2 Departures from the timeline

Situation	Do this	Why
The market gaps at the open	Execute anyway	The test's returns are open to open: gaps are already in its numbers
You miss the open	Execute at the next price you can, the same day; do not wait for a better one	Waiting for a better price is a new, untested rule

Situation	Do this	Why
A holiday on one market	Trade it at its next open	Each market keeps its own calendar
A partial fill, or a lot larger than the target	Round down to whole lots; leave a remainder flat	Rounding up adds risk the sizing did not ask for
Crypto, which never closes	Use the exchange's daily bar (UTC) as the close	That is how the test's crypto bars were built
The weight is below one lot	Skip the market, or use a smaller contract	A slice too small to trade is common with large lots

19.2 Sizing, with real numbers

AFTER THIS CHAPTER YOU CAN

- turn a CHART cell into a weight, a notional and a number of lots
- say how much risk the tested sizing actually takes

```
weight   = size × 10% ÷ annualised volatility,  capped at ±2       (size = the CHART cell's
notional = weight × the capital slice for that market                  (slice = capital ÷ nu
lots     = notional ÷ (price × lot size), rounded down
```

Volatility is the EWMA (span 60) of daily returns, annualised by √252. On TradingView, the built-in Historical Volatility indicator with length 60 is a close stand-in. The 10% is a *per-market* volatility target at full size. Because size is usually well below 1, most positions run far below it.

Real numbers, 30 October 2026. The six case-study markets, read from the reference engine at that close, on ₹10,00,000 split into six slices of ₹1,66,667 MEASURED:

Table 19.3 A six-market book on 30 October 2026

Market	CHART cell	Volatility (annual)	Target weight	Target notional
NIFTY 50	SHORT 0.20	12.9%	−0.158	−₹26,366
RELIANCE	SHORT 0.50	22.1%	−0.225	−₹37,460
Gold (USD)	FLAT · disagree	22.0%	0	₹0
USD/INR	LONG 0.10	8.6%	+0.117	₹19,576
BTC/USDT	FLAT · disagree	27.9%	0	₹0
BANKBEES	SHORT 0.16	13.2%	−0.123	−₹20,559

The gross exposure is about ₹1,03,962, roughly 10% of capital, with two of six markets flat. That is typical. The tested book's volatility at 10% per market was only about 1.3% to 1.6% a year, because sizes are small and markets diversify one another MEASURED. Scaled up to 10% for the whole book, the same paths would have drawn down roughly 16% to 29%. Decide which you are running before you start.

Lots

With NIFTY at 23,053, a single index-futures lot is worth several times a ₹1.67 lakh slice (check the exchange's current lot size). Small accounts cannot hold index futures at the tested size. Use the ETF (NIFTYBEES), fewer and larger slices, or accept that the book holds fewer markets. Fewer markets means more single-market noise (section 18.2). Every rounding should go down, never up.

19.3 Scaling in, adding and taking off

> **AFTER THIS CHAPTER YOU CAN**
>
> - know which scaling the system already does for you
> - add or reduce discretionarily without breaking the tested rules

Accepted momentum already scales. Its size is momentum's score, so as a trend strengthens over weeks the weekly rebalance adds, and as it weakens the rebalance takes off. Pyramiding and partial exits are built in. On the tested system, do nothing beyond the rebalance.

Discretionary traders who want to time inside the system's position, entering it or adding to it, have one measured tool and one measured caution MEASURED:

Table 19.4 Discretion inside the system

Moment	Action	Evidence
The CHART cell turns LONG or SHORT	Take the first part of the position at the next open	The system's own decision MEASURED
A close out of value in the trend's direction, AM agreeing	Take the next part	Breakouts from value beat the trend's average bar MEASURED
A pullback into value	Do not add because of it	No edge measured MEASURED
A probe against the position (126-bar trail)	Do not add; optionally take a step off	The side did worse than usual afterwards MEASURED
The next weekly rebalance	Bring the whole position to the tested target	Ends the discretion: you are back on the system

> **Never average down against the system.** Adding to a long that Accepted momentum has taken flat or short is a new strategy with no test behind it. The only measured buy-the-dip in Pragati is the investor's capitulation plan (section 23.1), which lives on the grid and at a different horizon.

19.4 Portfolio heat and correlation

> **AFTER THIS CHAPTER YOU CAN**
>
> - cap correlated exposure
> - set a drawdown ladder before the first trade

The tested portfolio gave every market equal risk. In a real account, markets come in clusters that move together: Indian banks, oil and gas, precious metals, INR pairs. Ten bank

stocks are not ten markets.

1. **Count clusters, not tickers**
   Group markets that move together. Sizing treats each cluster as at most two or three slices in total, however many names it holds UNMEASURED.

2. **Cap gross exposure**
   Set a ceiling on the sum of |weight| across the book, for example 1.5× capital, so that low-volatility markets at their cap of 2 cannot dominate UNMEASURED.

3. **Write the drawdown ladder now**
   At the backtest's worst drawdown, halve all weights. At 1.5× it, stop and review: the system's kill rule (section 21.3).

4. **Shorts need instruments**
   Shorting Indian cash stocks overnight needs futures. Without them, run long and flat only, and expect a different return profile from the test.

CHAPTER 20

The trading setups

20.1 How to read a setup card

AFTER THIS CHAPTER YOU CAN

- read any setup card in this book and know exactly how much to trust it

Every setup is written as a card. The fields are always the same, and none is optional.

Table 20.1 The setup card

Field	What it holds
Context	The regime and readings that must hold first (chapter 15)
Trigger	The closed-bar event that starts the setup
Entry	When and how the order is placed
Size	How large, in the tested formula or a stated fraction of it
Stop / invalidation	What proves the setup wrong, and where the risk is cut
Exit	What ends the trade when it works
Evidence	The tag, with numbers. **Measured** results name their verdict: confirmed, consistent, positive but unpredicted, or not supported
Fails when	The conditions under which it is known, or expected, to fail

Only T1 is a system. T1, Accepted momentum, passed a held-out test as a whole system. T2 to T4 are measured refinements of *when* to act inside it. T5 and T6 rest on the v9 audit's evidence for the grid's states, with trade rules that are a house plan. T7 is unmeasured structure. Size them in that order of trust.

20.2 T1 · Accepted momentum

T1 **Accepted momentum**	MEASURED	the tested system
Context	Any regime. The system is flat by design when the trail and momentum disagree	
Trigger	The CHART cell at the close: LONG x, SHORT x or FLAT	
Entry	Next open, on the weekly rebalance day, or at once on the first day of the book	
Size	x × 10% ÷ annualised volatility, capped at 2, per market, every market alike	
Stop / invalidation	None per trade: the side flips, or the size falls to flat. The book has a kill rule (section 21.3)	
Exit	When the CHART cell says FLAT or the opposite side, at the next rebalance	
Evidence	411 unseen markets: Sharpe +0.79 (t 4.0); 2020–26: +1.21 (t 3.7), net of costs. Timing beyond leaning long: t 4.1 and 2.2. Did not beat plain momentum	
Fails when	Run on one market (median Sharpe +0.23 to +0.38; a quarter lost); INR pairs; sharp reversals, which every trend system pays for	

20.3 T2 · Break of value with the trend

T2 **Break of value with the trend**	MEASURED	positive everywhere, unpredicted
Context	Accepted momentum LONG or SHORT (it agrees with the trail)	
Trigger	A close out of the value ribbon in the trend's direction, the bar before inside it	
Entry	Next open: the first part of a new position, or an add inside the system's target	
Size	Within the system's target weight; never above it	
Stop / invalidation	A close back inside value within a few bars is the first warning; the system's flat is the exit UNMEASURED	
Exit	With the system	
Evidence	+0.06σ over 10 bars beyond the trend's average bar: t 4.1 (DEV), 2.5 (2020–26), 10.4 (HA). Beat pullbacks into value everywhere. Hit rate 51–54%	
Fails when	In balance (not tested); when the trend is FLAT · disagree; as a standalone trade outside the system (not tested)	

20.4 T3 · The acceptance entry

T3 The acceptance entry MEASURED consistent

Context	A 126-bar trail flip: the trail has just changed side
Trigger	The flip bar (a ring on the chart; AUCTION WHY "accepted")
Entry	Next open, if Accepted momentum also turns that way. A flip with momentum disagreeing is FLAT
Size	The system's target
Stop / invalidation	The trail flipping back, which is rare at 126 bars
Exit	With the system
Evidence	+0.02 · +0.09 · +0.08σ over 10 bars (t 0.6 · 1.6 · 4.4); +0.05 · +0.11 · +0.11σ over 20. The predicted sign in all three sets, significant only in HA. 949 · 477 · 5,189 flips
Fails when	Flips are rare: on the six case markets, one every 1.3 to 2.2 years on the 126-bar trail. A single market gives few chances

20.5 T4 · The probe caution

T4 The probe caution MEASURED the opposite of fakeout-fading

Context	You hold a position on the trail's side
Trigger	A gold × against your side
Entry	None. This is a management card
Size	Hold the system's size. A discretionary trader may take one step off UNMEASURED
Stop / invalidation	The flip, if the probe is accepted; on the 126-bar trail fewer than 1 in 20 are
Exit	With the system
Evidence	After a probe the trail's side did worse than its average bar: −0.08σ (DEV), −0.05σ (2020–26), −0.07σ (HA, t −8.5) over 10 bars. Fading the probe measured nothing
Fails when	Read as an exit signal: it does not flip the side, and the system does not exit on it

20.6 T5 · The capitulation swing (long only)

T5 The capitulation swing (long only) EXTERNAL `state measured, trade rules a house plan`

Context	Not crypto. The grid in DOWN · cheap: Buy 4, green tint
Trigger	The ▲ at the close: value momentum reverting
Entry	Next open
Size	Small: a half or a third of a normal slice, since the measured lean is about +0.05σ over 10–20 bars UNMEASURED
Stop / invalidation	Below the capitulation's low, or 2 ATR, whichever is nearer UNMEASURED
Exit	After 20 bars (the measured horizon), or when the grid reaches BUILDING or PAID UNMEASURED
Evidence	Capitulation followed by gains in all three eras outside crypto (+0.06 to +0.08σ over 10–20 bars); the turn +0.046 to +0.056σ over 10 bars
Fails when	Crypto (the reading failed); a downtrend holding SHORT, where it is a counter-trend trade; one name at a time. The edge is across many events

20.7 T6 · The distribution avoid

T6 The distribution avoid EXTERNAL `state measured`

Context	The grid in DOWN · rich: Exit 0.25, red tint
Trigger	The ▼
Entry	No new longs. Investors trim toward a token weight
Size	—
Stop / invalidation	The grid leaving DOWN · rich toward the buyers
Exit	—
Evidence	As a state, followed by underperformance in every era; too rare to measure as an event
Fails when	Read as a short signal: shorting on the ▼ was never tested

20.8 T7 · Session-profile levels

T7 Session-profile levels UNMEASURED `reference structure`

Context	Intraday or swing, for traders who use profile levels already
Trigger	A test of VAH, VAL, the POC, VWAP, a naked POC or the IB edges
Entry	Your own rules
Size	Small, and never in conflict with the daily system's side
Stop / invalidation	Your own rules
Exit	Your own rules
Evidence	None in this repository. The naked-POC and VWAP "measured" labels are PVOI's and were not re-verified
Fails when	Costs below daily (section 10.2); trading against the daily side

20.9 When not to trade

> **AFTER THIS CHAPTER YOU CAN**
> - list the conditions under which no setup applies

- [ ] Any layer you need is paused, warming or grey (section 14.1).
- [ ] A non-standard chart.
- [ ] Accepted momentum reads FLAT · disagree, and you trade trends.
- [ ] The bar has not closed.
- [ ] Only one market would carry the risk of a portfolio's edge.
- [ ] Expiry week, if your read leans on OI.
- [ ] A crypto symbol, if your read leans on the capitulation state.
- [ ] You cannot say which setup card you are trading.

CHAPTER 21

Managing and reviewing

21.1 Discretionary overlays

These add context to a trade. None of them was tested as a rule UNMEASURED. Use them to adjust timing or size within the system's rules, not to replace them.

Table 21.1 Discretionary overlays

Overlay	Use it to	Caution
Grid family	Avoid adding to a long in PAID, STALLING or DISTRIBUTION; prefer adds in BASING or FADING	The grid is an investor's horizon
▲ during an AM short	Tighten or reduce the short: capitulation leans up for 10 to 20 bars	The system stays short by rule
OI character	A breakout on long build-up is committed; one on short covering is fragile	Builds did not predict continuation EXTERNAL
AUCTION probe	A probe against your side is a caution, not an exit: the side did worse than usual afterwards, but fewer than 1 in 20 flip the 126-bar trail MEASURED	Accepted breaks flip the trail at the close
Migration	Migrating with your side supports holding; Shifting against it is an early warning	Unmeasured
Quiet regime (gold)	Expect amplified readings; size down	A display state, not a direction

21.2 Stops, targets and invalidation

The system exits by its own rules. For a discretionary trader, Pragati's structure gives natural levels UNMEASURED.

Table 21.2 Structural levels for stops and targets

Level	Long trade	Short trade
The trail	The floor of accepted value. A close below it with the POC following and value migrating flips the trail	The ceiling, mirrored
VAL / VAH	Initial stop below the session VAL	Initial stop above the session VAH
POC	First target from below value; the balance point	First target from above value
Naked POC	An unfinished level above: a target	An unfinished level below: a target
Prior session VAH / VAL	Acceptance above the prior VAH supports the trade	Acceptance below the prior VAL supports it
VWAP	Holding above the session VWAP intraday	Holding below it

21.3 Exits, kill rules and re-entry

> **AFTER THIS CHAPTER YOU CAN**
>
> - write down every exit rule before the first entry

Table 21.3 Exits

Exit	Rule	Evidence
The system's exit	The CHART cell reads FLAT or the other side; act at the next rebalance	MEASURED
The kill rule	Stop the book if its drawdown passes 1.5× the backtest's worst, or if the rolling 12-month Sharpe stays below zero for two quarters	A rule written before results, not a measured edge
Time stop (T5)	20 bars after a capitulation entry	The measured horizon EXTERNAL
Structural stop (discretionary)	Beyond the swing extreme or the opposite value edge (section 21.2)	UNMEASURED
Probe against you	Not an exit (section 20.5)	MEASURED

After a kill. Stop trading the book. Paper-trade it again for at least six months at the same costs, then restart at a tenth of the intended size. A system stopped by its kill rule is not restarted because the next signal looks good.

21.4 The journal and the weekly review

> **AFTER THIS CHAPTER YOU CAN**
>
> - keep a journal that can tell skill from luck

Record every decision, including the decision not to act. The one-line reading from the routine (section 14.2) is the core of each entry.

Table 21.4 A journal entry

Field	Example
Date, market, timeframe	2026-10-30 · RELIANCE · D
One-line reading	readable · trend down (SHORT 0.50) · below value, no probe · Accumulate 1.5, sellers running, still cheapening · no OI · no trigger
Setup card	T1 (rebalance day)
Decision and size	Short, target −0.22 of the slice; rounded to 0 lots of futures, executed in the ETF
Execution	Next open, filled at …; slippage …
Expectation	The system's; no single-trade target
Outcome (filled later)	Return, days held, the readings at exit

Weekly review

- [ ] Did every action match a setup card? List the ones that did not.
- [ ] Book drawdown against the ladder (section 19.4).
- [ ] Costs and slippage against the test's assumptions (5 to 20 bps one way by class).
- [ ] The share of markets in each regime: are you trading trends in a book that is mostly balance?
- [ ] Any gold readings that you acted through?

PART VII

Investment guide

Weeks to months

22	Investing with the grid	85
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CHAPTER 22

Investing with the grid

22.1 The investor's frame

For investing, the grid is the tool. Each name in your universe sits in one family, and its units say how much weight it deserves relative to the others. You hold everything you would hold anyway, more of what is cheap with sellers exhausted, and less of what is rich with sellers in control. Units are relative weights for a book of many names, not a buy-or-sell switch for one.

> Evidence EXTERNAL: the graded units were positive in all three eras outside crypto. In Pragyam's allocator the units beat its seed weights in both eras on the Nifty 50 and the Dow 30.

Daily chart	Conviction on Ladder down, value on Ladder up: exactly what Pragyam sizes from.
Read at the week's last close UNMEASURED	One reading per name: the CVG row's STATE. cadence
Units to weights	Weight = units ÷ sum of units across the book.
Rebalance with a band UNMEASURED	Trade only where a weight moved by more than a tenth of its target. house rule

22.2 Nine families, nine actions

Table 22.1 The nine families as investment actions

Family	Condition	Action · units	Typical next family	What it says
DISLOCATED	Sellers in control, price cheap	Buy · 4	BASING, FADING, TURNED	The capitulation cell. The strongest measured lean. Stage in; add on the ▲
TURNED	Buyers in control, price cheap	Buy · 3	BUILDING	A turn while still cheap. Buy

Family	Condition	Action · units	Typical next family	What it says
BASING	Nobody in control, cheap	Accumulate · 1.5	TURNED, DISLOCATED	Accumulate slowly
FADING	Sellers in control, fair price	Accumulate · 1.5	DISLOCATED, IDLE	Sellers at a fair price were followed by gains (v8, outside crypto) EXTERNAL
BUILDING	Buyers in control, fair price	Hold · 1.5	PAID, IDLE	Hold. Buyers at a fair price lagged (v8), so do not add
IDLE	Nobody in control, fair	Wait · 1	anything	Do nothing
PAID	Buyers in control, rich	Trim · 0.75	STALLING, DISTRIBUTION	The move is paid for. Trim
STALLING	Nobody in control, rich	Trim · 0.75	DISTRIBUTION, IDLE	Trim
DISTRIBUTION	Sellers in control, rich	Exit · 0.25	FADING, STALLING	Exit toward a token weight. The ▼ marks the entry

CHAPTER 23

The investing setups

23.1 I1 · The capitulation plan

The one event the v9 audit found positive in every era is the capitulation turn. This is a staged plan around it. The staging itself is a house rule UNMEASURED; the state and the turn are EXTERNAL.

1 · Enter DOWN · cheap	The green tint appears; STATE reads Buy 4. Buy the first third.
2 · ▲ fires	Value momentum turns back toward fair; the tint deepens. Buy the second third.
3 · Leaves the cell	BASING or FADING (Accumulate 1.5), or TURNED (Buy 3). Buy the last third on the next push ↑.
4 · BUILDING	Hold 1.5. Let the position run.
5 · PAID or STALLING	Trim 0.75.
6 · ▼ DISTRIBUTION	Exit toward 0.25.

When the plan is wrong

- **Crypto.** The capitulation reading failed there. The tint and the ▲ are drawn gold. Skip the plan.
- **Gold readings.** A held row, a quiet regime, "basket settling", or a paused grid. Wait for them to clear before adding.
- **A trend against you.** If Accepted momentum is SHORT, the long-horizon trend still points down. Keep each tranche small (section 27.1).

23.2 I2 · The trend filter

I2 Accepted momentum as an investor's filter MEASURED

`not confirmed: drawdown halved, return halved`

Context	A long-only book of stocks, ETFs or indices
Rule	Hold a name while its CHART cell reads LONG; hold cash (or the rest of the book) while it reads FLAT or SHORT
Check	Weekly, at the close, and immediately when the state changes; trade the next open
Size	Unlevered: the name's ordinary weight, or its grid units (section 24.1)
Evidence	Below. Max drawdown smaller in all three sets; Sharpe about equal except 2020–26, where buy-and-hold's was 0.23 higher. Invested 43–53% of the time
Fails when	Long, strong bull markets (it sits out the recoveries' first legs); the cost is about half the compound return

Table 23.1 The filter against buy-and-hold (setup lab I1; equal-weight, daily-rebalanced portfolios)

Set	Markets	Invested	Filter: CAGR · max DD · Sharpe	Buy-and-hold: CAGR · max DD · Sharpe
DEV, 1992–2019	106	46%	11.5% · −31% · 0.82	20.0% · −58% · 0.79
HB, 2020–26	106	53%	7.4% · −29% · 0.81	18.2% · −36% · 1.04
HA, 391 NSE stocks, 1997–2026	391	43%	16.5% · −28% · 1.36	37.8% · −64% · 1.39

> **Read it as insurance, priced.** The filter roughly halves the deepest drawdown and roughly halves the return. It is a choice about which pain you can sit through. It is not a better portfolio. HA is today's NSE listings, so it carries survivorship; only the comparison on the same markets is meaningful. The 20-bar trail made the same trade, −35% against −73.5% drawdown at 13.5% against 24.4% growth (section 24.2).

23.3 I3 · Staging and averaging

I3 Entering in stages UNMEASURED a house rule

Context	Any name you intend to hold; the grid reads
Rule	Divide the intended weight into thirds. Each third needs a different, closed-bar reason
First third	The grid enters a Buy or Accumulate family (DISLOCATED, TURNED, BASING, FADING)
Second third	The ▲ (in DOWN · cheap), or the trend filter turning LONG, or a push ↑ that moves the row
Last third	The family improves (for example BASING to TURNED), or a weekly close confirms it
Averaging down	Allowed only inside the capitulation plan, and only while the grid stays in a Buy family. Never because the price is lower
Evidence	The families' units and the ▲ EXTERNAL; the thirds themselves are a house rule
Fails when	On crypto; in a held row; when a family is reached from above (TURNED falling to BASING is weakening, not a new entry)

23.4 I4 · Trimming and exiting

I4 Trim and exit discipline EXTERNAL states measured; the trims are units

Trim	PAID or STALLING: bring the name toward 0.75 units' share of the book
Exit	DISTRIBUTION (▼ marks its entry): toward 0.25 units, a token weight
Hold	BUILDING and IDLE change nothing. Do not add in BUILDING: buyers at a fair price lagged in v8
Speed	Trims are gradual: at each weekly review, move half the gap to the target UNMEASURED
Evidence	Distribution was followed by underperformance in every era; the graded units positive in all three eras outside crypto
Fails when	A trim is read as a short; taxes and costs are ignored. Small trims in a large book are often cheaper skipped

23.5 I5 · The rebalancing calendar

> **AFTER THIS CHAPTER YOU CAN**
>
> - run the investor's week in twenty minutes

1 **Friday's close (or the week's last close)**
Read every name's CVG STATE, units and WHY. Note ▲ ▼ and tint changes during the week.

2 **Compute weights**
Weight = units ÷ the sum of units, capped at about a tenth of the book per name (section 24.1).

3 **Move half the gap**
For each name, trade half the difference between current and target weight `UNMEASURED`. Skip trades smaller than your cost threshold.

4 **Apply the trend filter, if you use it**
Names whose CHART cell is not LONG go to cash, or keep only their capitulation tranches (section 23.2).

5 **Monday's open**
Execute.

6 **Monthly**
Review the drawdown ladder, the clusters and the watchlist.

23.6 I6 · Building a watchlist

> **AFTER THIS CHAPTER YOU CAN**
> - turn one indicator into a scan across many names

Pragati reads one chart at a time; it has no screener. A watchlist routine makes it scale.

1 **Choose a universe and keep it**
An index's members or a fixed list of ETFs. Changing the list because of recent winners is survivorship you add yourself.

2 **One alert per name**
Create Pragati's `Any alert() function call` alert on each chart with `Alert on capitulation state entry / exit` on. Every ▲ ▼ and every capitulation entry and exit arrives with the ticker, the grid's action and units, and both tapes.

3 **A weekly sweep**
Open each name on the weekly chart once a week and log its one-line reading (section 14.2).

4 **Rank by units**
Highest units first. That is the order the grid would allocate in.

5 **Note the clusters**
Five capitulations in one sector are one bet.

CHAPTER 24

The book and its risk

24.1 Building the book: a worked example

Six names, read at the week's close. Weights are units divided by their sum.

Table 24.1 A six-name book

Name	CVG STATE	Units	Weight
A	`Buy 3.6` · DISLOCATED	3.6	32.4%
B	`Buy 2.8` · TURNED	2.8	25.2%
C	`Hold 1.5` · BUILDING	1.5	13.5%
D	`Accumulate 1.4` · BASING	1.4	12.6%
E	`Wait 1.0` · IDLE	1.0	9.0%
F	`Trim 0.8` · PAID	0.8	7.2%
Total		11.1	100% (rounded)

With a small book, one name can grow large. A practical cap of about a tenth of the book per name keeps the plan diversified UNMEASURED. The more names, the closer you get to how the units were measured.

24.2 Investor risk

- **Trend filters trade return for drawdown.** On 119 markets, holding only while the 20-bar trail was up cut the worst drawdown from −73.5% to −35%, but gave up about 11 points of yearly growth (13.5% against 24.4%) MEASURED. The 200-day average and plain momentum made the same trade. Accepted momentum used the same way, as a long-only filter, halved the drawdown in all three sets of the setup lab at about half the return (section 23.2) MEASURED.
- **The ▲ is a lean.** About +0.05σ over 10 to 20 bars. It pays across many names and many events, not on any one EXTERNAL.
- **Gold is caution.** Reduce or wait; never read gold as a direction.

PART VIII

Case studies

Real and constructed

25	Real markets, reconstructed	95
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CHAPTER 25

Real markets, reconstructed

25.1 How the cases were chosen

> **AFTER THIS CHAPTER YOU CAN**
>
> - know exactly what a real-data case shows, and what it cannot

These six cases are real daily bars, run through the repository's reference implementation of the auction engine, the push gate and Accepted momentum. The reference is the same code as the setup lab, and it was parity-tested against the Pine. They are not TradingView screenshots, and they show only what the reference computes:

- the 126-bar trail, its value area and POC;
- probes and flips;
- migration;
- S and the position;
- the weight the tested sizing would have traded.

They do not show the tapes, the grid or the ▲▼, which need TradingView's intrabar data and Samanvaya's macro drivers. Chapter 26's walkthroughs, which are constructed, cover those.

Table 25.1 The selection rule

Rule	Fixed in advance (docs/setup-lab.md §5)
Markets	One per class: NIFTY 50 (index), RELIANCE (NSE stock), gold (commodity), USD/INR (FX), BTC/USDT (crypto), BANKBEES (ETF)
Window	The last 300 daily bars to the data's end, 30 October 2026 (31 October for BTC, which trades daily)
Events	The most recent event of each type before 30 September 2026 with a complete 20-bar outcome, whatever the outcome
Campaigns	Every campaign of the traded weight inside the window, winners and losers

So the cases were not picked for looking good. Several lose. The plates draw the trail as Pragati does with Accepted momentum on: hue is the side, intensity the size, grey when flat. Below each plate is the weight the tested sizing would have held. Letters mark the dates the text discusses. Event outcomes are in σ units, as in the setup lab: the move over 10 and 20 bars from the next open, in the event's direction.

25.2 NIFTY 50: a floor that never moved

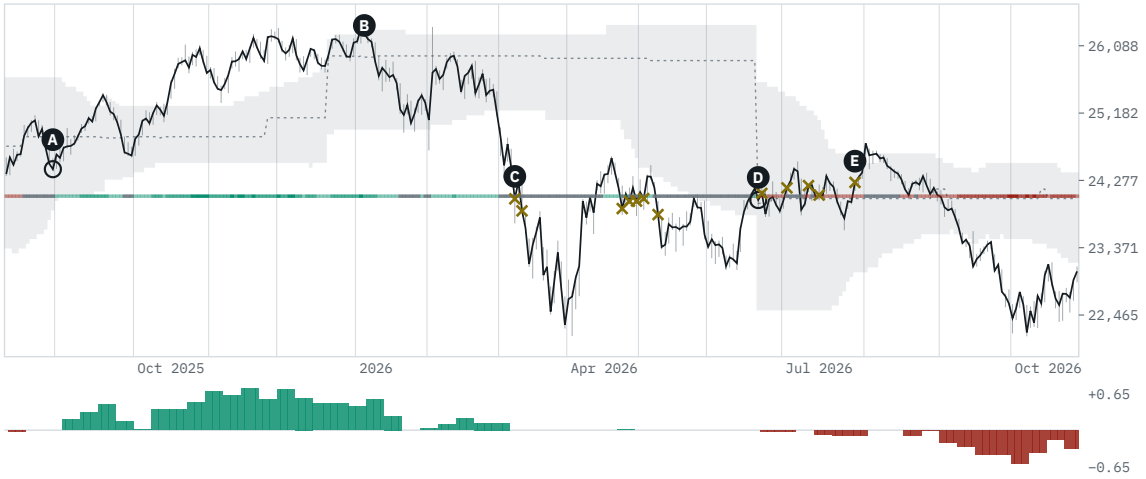


Figure 25.1 NIFTY 50, daily, 2025-08-08 to 2026-10-30, reconstructed by the reference engine MEASURED. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

A · 29 August 2025. The trail flips up at 24,064, with the close at 24,427. Value had shifted up (migration +1), and the window's POC, 24,864, stood above the old ceiling: acceptance. Momentum's S was −0.01, so the CHART cell read `FLAT · disagree` on the flip itself. The long campaign began only at the 3 September rebalance, once S turned up.

The floor never stepped. NIFTY rallied to a closing high of 26,329 on 2 January 2026, and the trail stayed at 24,064 throughout. Conviction's quiet regime held on 85% of these 300 bars, the push gate was shut, and a trail can step only through an open gate (section 8.2). The position did not care. With Accepted momentum on, the side and S decide, and the traded weight reached 0.65 during the rally.

B · 5 January 2026. The latest pullback into value before the cut-off, a setup-lab S3 event. NIFTY closed back inside value at 26,250, and over the next 10 bars it fell 2.67σ further. The pullback was not a buy, as the setup lab found on average (section 16.4).

C · 9 March 2026. The first of seven probes below the floor. Within 20 bars NIFTY traded down to 22,183, 8% below the trail, and still **did not flip**. Migration stayed +1 (value six months earlier was lower still), so the break was never accepted. S had turned to −0.33, so the system was already FLAT. That is the trail working as designed: acceptance is about value, not distance.

D · 19 June 2026. The flip down came with price almost on the line: close 24,013 against the trail's 24,064. The POC (23,968) had moved below and migration read −1. The level did not move; the broken floor became the ceiling.

E · 29 July 2026. The latest of five probes *above* the new ceiling, at 24,250. It was not accepted, and the market drifted (−0.28σ over 10 bars in the trail's direction). The short cam-

paign began at the 18 August rebalance and was still open at the data's end, having made +0.55% on the slice while NIFTY fell 4.9%.

Table 25.2 NIFTY 50: every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-08-11 → 2025-08-18	SHORT	5	0.02	+1.6%	−0.04%
2025-09-03 → 2026-01-20	LONG	95	0.65	+1.5%	−0.24%
2026-01-29 → 2026-03-05	LONG	25	0.18	−5.5%	−0.77%
2026-04-23 → 2026-04-29	LONG	5	0.01	−0.2%	−0.00%
2026-06-22 → 2026-07-06	SHORT	10	0.02	+0.8%	−0.02%
2026-07-14 → 2026-08-03	SHORT	15	0.09	+2.4%	−0.21%
2026-08-18 → 2026-10-30 (open)	SHORT	49	0.53	−4.9%	+0.55%

Table 25.3 NIFTY 50: the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest rejected probe	2026-07-29	down	24,250 vs 24,064	−0.28σ	−0.03σ	—
Latest pullback into value (S3)	2026-01-05	up	26,250 vs 24,064	−2.67σ	−0.77σ	—
Latest accepted probe	2025-04-25	up	24,039 vs 24,064	+0.96σ	+0.76σ	2025-05-09

At the data's end

Close 23,053	Trail down 24,064	Value area 23,168 – 24,384
Location below value	AUCTION STATE Migrating ↓	S −0.20
CHART cell SHORT 0.20	Volatility 12.9% a year	Target weight −0.158

> **What it teaches.** A six-month trail flips rarely, and never on distance alone. A frozen floor is not a risk measure. The system's size, not the line, carried the position.

25.3 RELIANCE: right side, wrong timing

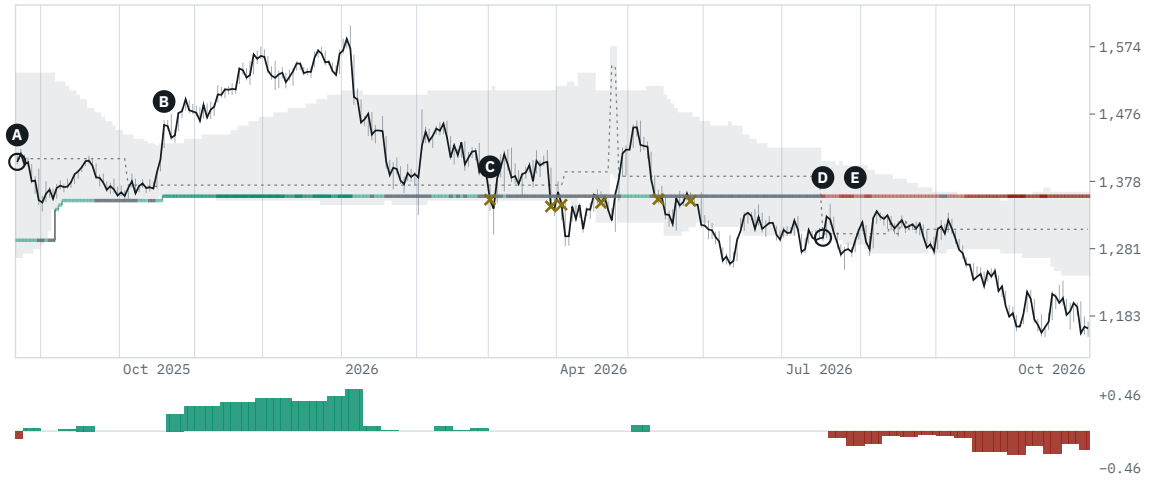


Figure 25.2 RELIANCE, daily, 2025-08-20 to 2026-10-30, reconstructed by the reference engine `MEASURED`. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

A · 20 August 2025. The trail flips up at 1,292.8 (close 1,406.5), and the CHART cell turns `LONG 0.11`. Yet the traded weight that day was still −0.09, from the earlier short. The traded position waits for the next weekly rebalance. That is the tested rule, and it is why the plate's weight strip lags the trail's colour.

B · 20 October 2025. The fourth step since early September, to 1,356.8. RELIANCE's gate was open far more often than NIFTY's (quiet on only 13% of these bars), so its trail ratcheted up behind the move.

The long campaign lost. It ran 65 bars from 21 October 2025 to 22 January 2026, and its size peaked as price did: the stock's closing high was 1,585 on 2 January. Net, it made −1.41% on the slice, while the stock moved −6.4% open to open. Momentum is strongest at the top of a move. That is the cost every trend system pays, and the reason the edge is measured on hundreds of markets.

C · 2 March 2026. The first of six probes below the floor (close 1,351.75 against 1,356.8), with migration 0. The probes came and went for four months without acceptance.

D · 16 July 2026. Acceptance: a close of 1,296.6, the POC at 1,302.4 below the trail and migration −1. The short campaign began at the 20 July rebalance.

E · 30 July 2026. A pullback into value on the short side (close 1,292.9). It did nothing measurable: −0.43σ against the trade over 10 bars, +0.14σ over 20. By 30 October the stock was at 1,165, below the value area. The open short campaign had made +0.56% on the slice while the stock fell 11.4%.

Table 25.4 RELIANCE: every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-08-20 → 2025-08-21	SHORT	2	0.09	−0.7%	+0.05%
2025-08-22 → 2025-08-29	LONG	5	0.03	−4.2%	−0.16%
2025-09-08 → 2025-09-19	LONG	10	0.06	+1.0%	−0.03%

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-10-21 → 2026-01-22	LONG	65	0.46	−6.4%	−1.41%
2026-02-09 → 2026-02-27	LONG	15	0.05	−9.6%	−0.33%
2026-05-04 → 2026-05-08	LONG	5	0.07	−4.7%	−0.33%
2026-07-20 → 2026-10-30 (open)	SHORT	73	0.26	−11.4%	+0.56%

Table 25.5 RELIANCE: the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest pullback into value (S3)	2026-07-30	down	1,293 vs 1,357	−0.43σ	+0.14σ	—
Latest rejected probe	2026-05-26	up	1,350 vs 1,357	−1.11σ	−0.45σ	—
Latest accepted probe	2023-02-22	up	1,081 vs 1,101	−0.40σ	−1.05σ	2023-03-10

At the data's end

Close 1,165	Trail down 1,357	Value area 1,242 – 1,363
Location below value	AUCTION STATE Shifting ↓	S −0.50
CHART cell SHORT 0.50	Volatility 22.1% a year	Target weight −0.225

> **What it teaches.** The traded weight lags the CHART cell by up to a week. Being on the right side is not the same as making money on one market. A long probe series can precede acceptance by months.

25.4 Gold: the size leaves before the line

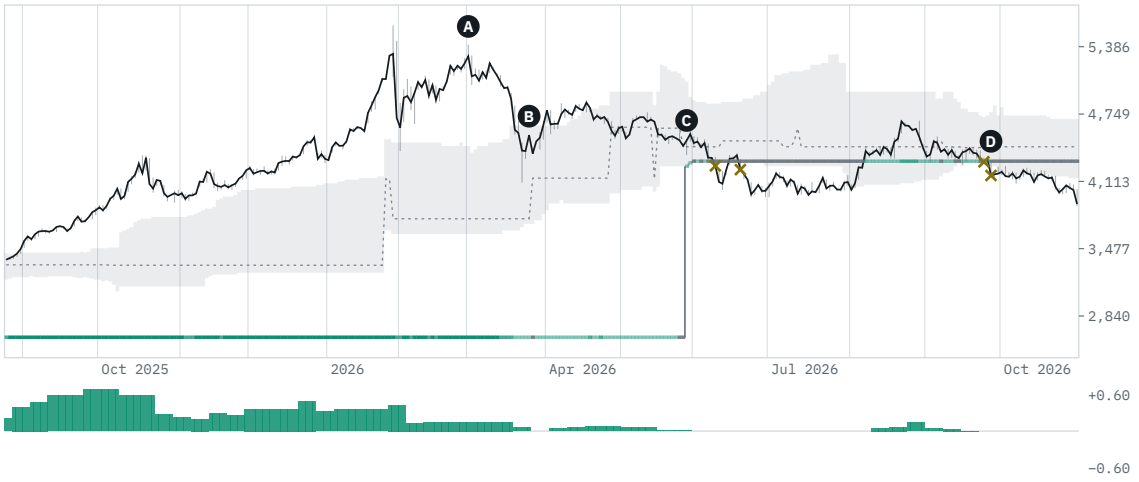


Figure 25.3 Gold (USD), daily, 2025-08-25 to 2026-10-30, reconstructed by the reference engine MEASURED. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

The best campaign of the six. Gold's long ran 147 bars, from the window's start on 25 August 2025 to 25 March 2026. The weight reached 0.60, and the campaign made +14.5% on the slice while gold rose 32.9% open to open.

A · 2 March 2026. Gold closed at 5,294, near its closing high of 5,318 on 29 January, with S at +0.56. Over the next 20 bars it traded down to 4,101.

B · 25 March 2026. S had fallen to +0.09, and the weight with it, to 0.05. The campaign ended by **size**, not by the trail, which was then at 2,640: 42% below price. A stop at the trail would have given back almost everything. The system did not wait for it.

C · 28 May 2026. The trail finally stepped, from 2,640 to 4,256, the first bar in months with the gate open and value migrated. It stepped twice more within days, to 4,304.

D · 28 September 2026. The latest probe, at 4,168, below the floor of 4,304. It was not accepted within 20 bars; gold drifted to 4,038 (−0.41σ over 20 bars against the trail's side, as the setup lab's F1 expects). At the data's end gold closed at 3,899, 9% below a trail that still reads up. The POC was 4,440 and migration 0, so the market had not accepted the lower prices. S was −0.32, so the system was flat.

Table 25.6 Gold (USD): every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-08-25 → 2026-03-25	LONG	147	0.60	+32.9%	+14.52%
2026-04-02 → 2026-05-29	LONG	40	0.07	−3.6%	−0.22%
2026-08-11 → 2026-09-22	LONG	30	0.12	−1.9%	−0.47%

Table 25.7 Gold (USD): the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest rejected probe	2026-09-28	up	4,168 vs 4,304	+0.22σ	−0.41σ	—
Latest pullback into value (S3)	2026-04-21	up	4,698 vs 2,640	−0.28σ	−0.56σ	—
Latest accepted probe	2022-08-26	up	1,736 vs 1,744	−0.27σ	−1.65σ	2022-08-31

At the data's end

Close	Trail	Value area
3,899	up 4,304	4,144 – 4,704
Location	**AUCTION STATE**	**S**
below value	Balance	−0.32
CHART cell	**Volatility**	**Target weight**
FLAT · disagree	22.0% a year	0

> **What it teaches.** Accepted momentum exits through S long before the trail is touched. A trail far from price is common, and is not a stop. A price below an up-trail with no acceptance is a probe: caution, and the system flat.

25.5 USD/INR: a staircase, and a quiet market's large weight

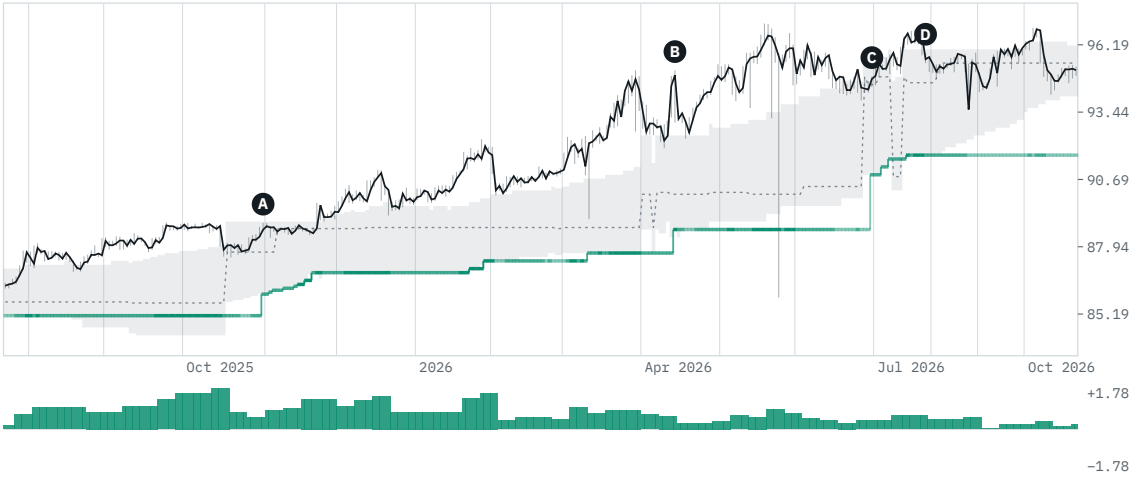


Figure 25.4 USD/INR, daily, 2025-07-23 to 2026-10-30, reconstructed by the reference engine MEASURED. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

A textbook managed trend. In 300 bars there were no probes and no flips, and sixteen steps. The trail ratcheted from 85.12 to 91.68 as USD/INR climbed from 86.3 to 95.2. Migration read Shifting or Migrating ↑ on 267 of the 300 bars.

A · 31 October 2025. The first step, 85.12 to 86.00. **B · 14 April 2026.** A step to 88.64. **C · 30 June 2026.** A step to 90.88.

The weight. Annualised volatility was only 4.6% at the window's start, so even modest sizes asked for large weights. The weight reached 1.78, close to the cap of 2. One campaign held the whole window and was still open. Yet it made only +2.1% on the slice against a +10% move. The weight happened to be high on the days the rupee strengthened and low on the days it weakened. In the first quarter of 2026 the pair rose 3.0% and the position made almost nothing. A trend the trail followed perfectly still did not pay much.

D · 29 July 2026. The latest pullback into value, at 95.60: +0.01σ over 10 bars, +0.14σ over 20. Nothing to see, as on average.

Table 25.8 USD/INR: every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-07-23 → 2026-10-30 (open)	LONG	300	1.78	+10.1%	+2.11%

Table 25.9 USD/INR: the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest pullback into value (S3)	2026-07-29	up	95.60 vs 91.68	+0.01σ	+0.14σ	—
Latest rejected probe	2023-07-25	up	81.81 vs 81.84	+1.83σ	+1.62σ	—
Latest accepted probe	2019-06-24	up	69.57 vs 69.76	−0.79σ	−0.08σ	2019-06-28

At the data's end

Close 95.15	Trail up 91.68	Value area 94.08 – 96.16
Location in value	AUCTION STATE Migrating ↑	S +0.10
CHART cell LONG 0.10	Volatility 8.6% a year	Target weight +0.117

> **What it teaches.** Low volatility means large weights: the cap and the gross ceiling matter (section 19.4). A perfect-looking trend can return little. The INR pairs lost money in the full test, whatever this window shows.

25.6 BTC/USDT: a falling ceiling and a transition

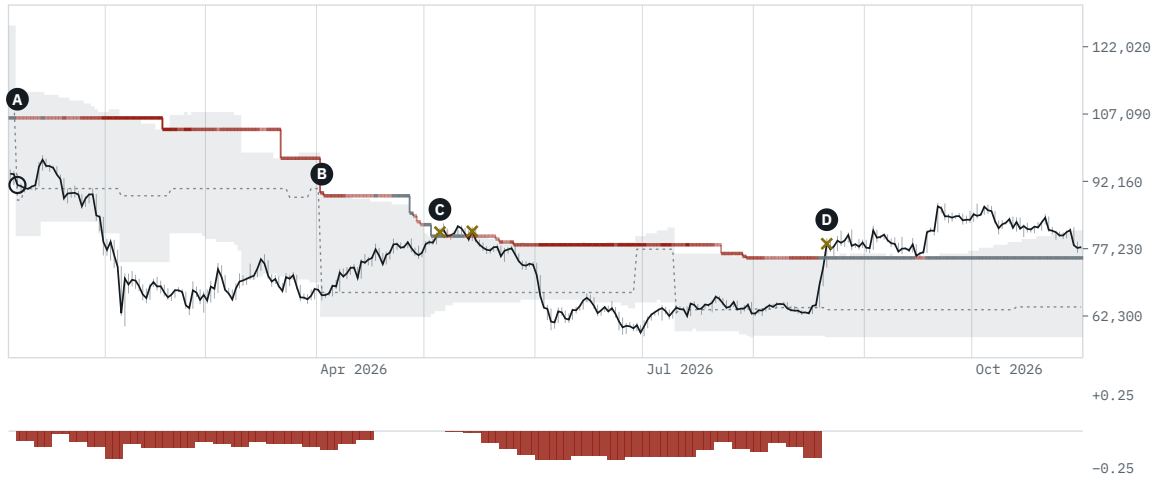


Figure 25.5 BTC/USDT, daily, 2026-01-05 to 2026-10-31, reconstructed by the reference engine MEASURED. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

A · 7 January 2026. The trail flips down at 106,240 (close 91,364): the POC at 88,000 had dropped below the old floor and migration read −1. Crypto trades every day, so these 300 bars span only ten months.

A ceiling that stepped. Crypto's push gate was open far more often (quiet on 14% of these bars), and the trail stepped down fifteen times, from 106,240 to 75,200, as BTC fell to a closing low of 58,625 on 30 June. **B · 2 April 2026:** one of those bars, with S at −0.48 and the trail stepping to 89,600.

Two short campaigns, both profitable but small. From 7 January to 16 April (100 bars) the system made +1.04% on the slice while BTC fell 15.6%. From 7 May to 19 August (105 bars) it made +0.17% on a fall of 8.7%. BTC's volatility (about 28% to 29% a year on this measure) kept the weight at 0.17 or below. High-volatility markets carry small weights by design.

C · 5 May 2026. A probe above the ceiling at 80,906. S had turned +0.07, so the cell read FLAT. The probe was not accepted, and the ceiling stepped lower again within weeks.

D · 21 August 2026. The latest probe above the ceiling (78,338 against 75,200), with S at +0.35. The trail is down and momentum up: `FLAT · disagree`, the transition regime (section 15.1). It was not accepted within 20 bars, and at the data's end BTC still sat above a down-trail in Balance, flat.

Table 25.10 BTC/USDT: every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2026-01-07 → 2026-04-16	SHORT	100	0.17	−15.6%	+1.04%
2026-05-07 → 2026-08-19	SHORT	105	0.17	−8.7%	+0.17%

Table 25.11 BTC/USDT: the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest rejected probe	2026-08-21	down	78,338 vs 75,200	−0.04σ	+0.21σ	—
Latest pullback into value (S3)	2026-07-10	down	64,162 vs 78,080	−0.28σ	−0.11σ	—
Latest accepted probe	2024-10-01	up	60,806 vs 62,080	+0.35σ	+0.91σ	2024-10-02

At the data's end

Close	Trail	Value area
77,582	down 75,200	57,600 – 81,280
Location	**AUCTION STATE**	**S**
in value	Balance	+0.11
CHART cell	**Volatility**	**Target weight**
FLAT · disagree	27.9% a year	0

> **What it teaches.** Volatility sets the size, and crypto's is small. FLAT · disagree is a regime, not an error. No capitulation reading applies to crypto (section 15.3).

25.7 BANKBEES: a crash that was never accepted

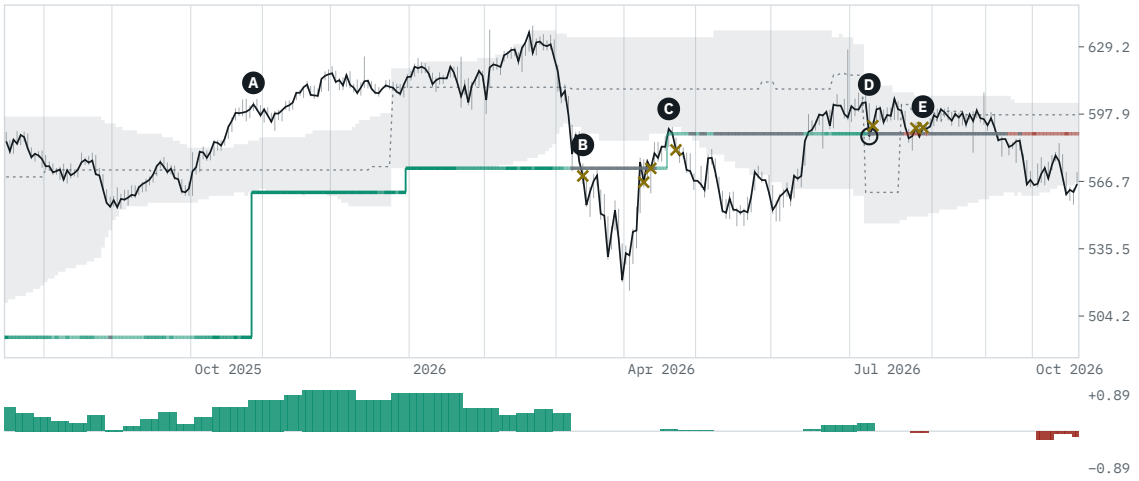


Figure 25.6 BANKBEES, daily, 2025-07-17 to 2026-10-30, reconstructed by the reference engine MEASURED. Above: close and range; the rolling 126-bar value area (band) and POC (dotted); the trail (hue the side, intensity the size, grey when flat); a gold × at each probe's first bar; a ring at each flip. Below: the weight the tested sizing traded.

A · 29 October 2025. After a long freeze, the gate opened with value migrated, and the trail jumped from 494.4 to 561.6 in one step. A trail that cannot step for months moves all at once when it can.

The long campaign ran from the window's start to 6 March 2026 (158 bars), with the weight up to 0.89. It made +0.22% on the slice for a +0.5% move.

B · 12 March 2026. A probe below the floor of 572.8. Within 20 bars the ETF fell to 516, 10% below the trail, and the break was still **not accepted**: migration stayed +1. It recovered to 591 by 21 April.

C · 21 April 2026. A step to 588.8. **D · 8 July 2026.** The flip down at 588.8, with the close at 587.7 and the POC at 561.6 below. Again the flip came with price at the line, not 10% beyond it.

E · 29 July 2026. The latest probe above the new ceiling (591.7). It was not accepted, and S was near zero, so the system stayed flat. By 30 October the ETF closed at 565.5, just below the value area's low edge, `SHORT 0.16`.

Table 25.12 BANKBEES: every campaign in the window

Campaign	Side	Bars	Largest weight	Price, open to open	Return on the slice, net
2025-07-17 → 2026-03-06	LONG	158	0.89	+0.5%	+0.22%
2026-04-17 → 2026-05-07	LONG	15	0.03	−2.4%	−0.04%
2026-06-12 → 2026-07-09	LONG	20	0.17	+1.3%	+0.10%
2026-07-24 → 2026-07-30	SHORT	5	0.03	+0.4%	−0.01%
2026-10-05 → 2026-10-30 (open)	SHORT	12	0.18	−1.2%	−0.06%

Table 25.13 BANKBEES: the most recent event of each type, whatever its outcome

Event (selection rule)	Date	Trail	Close vs trail	10 bars	20 bars	Flip
Latest rejected probe	2026-07-29	down	591.7 vs 588.8	−0.10σ	+0.22σ	—
Latest pullback into value (S3)	2026-02-19	up	626.3 vs 572.8	−2.70σ	−3.88σ	—
Latest accepted probe	2012-07-03	down	106.8 vs 105.6	−0.15σ	+0.14σ	2012-07-30

At the data's end

Close 565.5	Trail down 588.8	Value area 566.4 – 603.2
Location below value	AUCTION STATE Balance	S −0.16
CHART cell SHORT 0.16	Volatility 13.2% a year	Target weight −0.123

> **What it teaches.** Crashes are not acceptance. Flips happen when value moves, often quietly. A frozen trail that unfreezes jumps.

25.8 The six as one book

AFTER THIS CHAPTER YOU CAN

- see what the tested edge looks like on six markets rather than four hundred

Run as one book, with equal slices from 6 January 2026 (the common start of the six windows) to 30 October 2026, net of the test's costs, the six markets made **−0.69%**. The worst drawdown was −1.29%, at an annualised volatility of 1.6% MEASURED.

Table 25.14 Six markets, 6 January to 30 October 2026

Market	Contribution on its slice
NIFTY 50	−1.72%
RELIANCE	−3.10%
Gold	+1.76%
USD/INR	+0.12%
BTC/USDT	+1.17%
BANKBEES	−2.71%

That is a losing ten months, and it was chosen by rule, not for effect. It is also what the evidence predicts for a book this small:

- on one market the median Sharpe was +0.23 to +0.38, and a quarter of markets lost over the whole test;
- the edge appeared across 119 and 411 markets, sized alike;
- the book's volatility, 1.6%, is the tested book's own (1.3% to 1.6%), which is why its losses are small too.

A trader who scales the book to 10% volatility should expect this kind of stretch to look six times larger.

CHAPTER 26

Full-stack walkthroughs

26.1 Four worked scenarios

Illustrative sequences on a daily chart, not real cases. The combined readings are reasoning `UNMEASURED`; each book follows its own tool.

1 · A capitulation dip inside an uptrend

Table 26.1 A capitulation dip inside an uptrend

When	The readout says	Trader	Investor
Day 0	`LONG 0.58` · CVG `Hold 1.5` BUILDING	Long at system size	Hold 1.5
Day 6	CVG `Buy 3.9` DISLOCATED · sellers running · still cheapening · green tint · OI `Long unwinding` · `probe, not accepted`	Still long; the size follows S at the weekly rebalance	Buy the first third
Day 9	▲ · CVG `Buy 4.0` · turning up · the tint deepens	Long	Buy the second third
Day 14	CVG `Accumulate 1.4` BASING · PUSH ↑	Long	Buy the last third on the push
Day 40	CVG `Trim 0.8` PAID	Long, sized by S	Trim

2 · Distribution while the trend is still long

Table 26.2 Distribution while the trend is still long

When	The readout says	Trader	Investor
Day 0	`LONG 0.35` · Shifting ↓ · CVG `Trim 0.8` STALLING	Long; no adds	Trim
Day 3	▼ · CVG `Exit 0.27` DISTRIBUTION · OI `Short build-up`	Long until the system's rebalance says otherwise	Exit toward 0.25
Day 8	"accepted, flips at the close" · then `FLAT · disagree`	Flat at the next rebalance	Hold the token weight
Day 20	`SHORT 0.30` · Migrating ↓	Short	Token weight

3 · Nothing to do

CVG `Wait 1.0` IDLE · no edge; AUCTION `FLAT · disagree` · Balance; OI `Balanced`. The trader is flat and the investor waits. Most days look like this, and doing nothing is the instruction.

4 · Aligned up, with a crowded book

AUCTION `LONG 0.71` · Migrating ↑; CVG `Hold 1.5` BUILDING · buyers running; OI `Long build-up` · `book 88 pct`. The trader holds the system's size. The investor holds and does not chase: buyers at a fair price lagged (v8), and the gold book says the trade is crowded.

26.2 Three more walkthroughs (constructed)

AFTER THIS CHAPTER YOU CAN

- follow the whole stack, grid and OI included, through a sequence of closes

Constructed, not real. These readouts were written to teach the full stack, including the tapes, the grid and OI, which chapter 25's reconstructions cannot show. No market produced them, and no outcome is implied.

5 · A break of value that the grid confirms

Table 26.3 Walkthrough 5

Close	The readout says	Trader	Investor
Day 1	AUCTION Shifting ↑ · trail ↑ · in value · `LONG 0.31` · CVG `Hold 1.42` BUILDING · buyers running	Hold the target	Hold
Day 4	Close above VAH: out of value, its way · `LONG 0.36` · PUSH ↑↑	T2: take the next part, within the target	Hold; do not add in BUILDING
Day 9	CVG `Trim 0.81` PAID · value tape past +θ · trail line still green	Hold: the system sizes by S	Trim toward 0.75 units (I4)
Day 15	Probe × below the trail · migration still +2 · `LONG 0.22`	Hold; take a step off if discretionary (T4)	Hold the trimmed weight

6 · A capitulation on crypto

Table 26.4 Walkthrough 6

Close	The readout says	Reading
Day 1	CVG `Buy 3.6` DISLOCATED · pane tint **gold**, not green · `SHORT 0.40`	Crypto: the capitulation reading failed there. The gold tint says so
Day 3	▲ drawn in gold · value momentum reverting	Do not buy it as a capitulation (section 15.3)
Day 8	Trail ↓ · probe × above the ceiling · S turning up → `FLAT · disagree`	Transition. The trader is flat by the system; nothing to do

7 · Crowded and paid for

Table 26.5 Walkthrough 7

Close	The readout says	Reading
Day 1	OI `Long build-up` · book 84 pct (gold) · CVG `Trim 0.75` PAID · `LONG 0.58`	Committed buyers, a full book and a rich price: late, not wrong
Day 2	OI `Short covering` · histogram columns cast gold	The push is paid for by exits: pushes like this lagged `EXTERNAL`. Size down a step if discretionary
Day 6	▼ DISTRIBUTION · CVG `Exit 0.25` · trail still ↑ · `LONG 0.41`	Matrix A's conflict (section 27.1): the investor exits toward a token weight; the trader holds the system until it flips

PART IX

Permutations and combinations

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CHAPTER 27

The four matrices

27.1 Matrix A · The system against the grid: 27 cases

Rows are the nine grid families; columns are Accepted momentum's three positions. In each case, TRADE follows the tested system MEASURED and INVEST follows the family's action EXTERNAL. The combined reading is reasoning UNMEASURED. When they conflict, each book follows its own tool (section 28.1).

Table 27.1 The system against the grid, all 27 cases

Grid family	AM LONG	AM FLAT	AM SHORT
UP · cheap TURNED Buy 3	**Aligned. Trend, buyers and a cheap price agree: the strongest constructive picture.** TRADE Hold the system's size. INVEST Buy toward 3 units.	**Buyers are early; long momentum has not caught up.** TRADE No position: the system is flat. INVEST Buy toward 3 units. The grid leads here.	**A turn against a falling trend: buyers take control while price is still cheap.** TRADE The system stays short until its rules flip it. Expect a conflict. INVEST Build in stages. Buy toward 3 units.
UP · fair BUILDING Hold 1.5	**A healthy trend at a fair price.** TRADE Hold. INVEST Hold 1.5. Do not chase.	**Buyers in control, no trend yet.** TRADE Flat. INVEST Hold.	**A bounce inside a downtrend.** TRADE Short, at the system's size. INVEST Hold. No adds.
UP · rich PAID Trim 0.75	**A late trend: the move is already paid for.** TRADE Hold the system's size; it sizes by S. Do not add. INVEST Trim toward 0.75.	**Stretched, with no trend behind it.** TRADE Flat. INVEST Trim.	**A squeeze into a downtrend, at a rich price.** TRADE Short. INVEST Trim or avoid.
FAINT · cheap BASING Accumulate 1.5	**A pullback inside an uptrend, now cheap.** TRADE Hold. INVEST Accumulate.	**Basing: cheap, nobody in control.** TRADE Flat. INVEST Accumulate slowly.	**Cheap inside a downtrend, nobody in control.** TRADE Short. INVEST Accumulate small. Wait for DOWN · cheap or a turn.
FAINT · fair IDLE Wait 1	**The trend is resting.** TRADE Hold. INVEST Wait. Hold what you have.	**Nothing to read.** TRADE Flat. INVEST Wait.	**The downtrend is resting.** TRADE Short. INVEST Wait.

Grid family	AM LONG	AM FLAT	AM SHORT
FAINT · rich STALLING `Trim 0.75`	**An uptrend stalling at a rich price.** TRADE Hold. Watch for the system to go flat. INVEST Trim.	**Rich and stalling.** TRADE Flat. INVEST Trim.	**Weak, rich, falling.** TRADE Short. INVEST Trim or avoid.
DOWN · cheap DISLOCATED `Buy 4 · ▲ here`	**A capitulation dip inside a long uptrend.** TRADE Long holds; its size falls as S falls. INVEST Buy toward 4 units. Add on the ▲.	**Capitulation, trend undecided.** TRADE Flat. INVEST Stage in. Add on the ▲.	**Capitulation inside a downtrend: the classic conflict between horizons.** TRADE Short, by the trend rule. INVEST Stage in small. The ▲ is the trigger. Not on crypto.
DOWN · fair FADING `Accumulate 1.5`	**Sellers inside an uptrend, fair price.** TRADE Hold. INVEST Accumulate.	**Fading.** TRADE Flat. INVEST Accumulate slowly.	**A downtrend with sellers in control.** TRADE Short. INVEST Accumulate small.
DOWN · rich DISTRIBUTION `Exit 0.25 · ▼ here`	**Sellers have taken a rich price while the long trend is still up: early distribution.** TRADE Long until the system flips. Its size will fade. INVEST Exit toward 0.25.	**Distribution.** TRADE Flat. INVEST Exit.	**Distribution inside a downtrend: aligned to the downside.** TRADE Short. INVEST Exit.

27.2 Matrix B · Open interest against location in value

The OI row's character, read against where the close sits in the session profile (the panel's Location row, or the price against VAH and VAL). Practitioner readings UNMEASURED, except where tagged.

Table 27.2 Open interest against location in value

OI character	Above value	In value	Below value
`Long build-up`	New longs pushing for acceptance higher. Watch for the trail to accept	New longs inside balance	Longs adding below value: bottom-fishing with commitment
`Short build-up`	Shorts leaning against a breakout	New shorts inside balance	A breakdown with commitment
`Short covering`	A rally paid for by exits beyond value: fragile. Short-covering rallies lagged EXTERNAL	Covering inside balance	A relief bounce from below value
`Long unwinding`	Longs leaving up high: a distribution risk	Liquidation inside balance	Liquidation below value: fuel for a capitulation
`Balanced · Price flat`	No read	No read	No read

27.3 Matrix C · Trail side against migration

Table 27.3 Trail side against migration

Migration (AUCTION STATE)	Trail ↑	Trail ↓
`Migrating ↑`	Value moving with the trail. Steps are likely	Value has left upward. A close above with the POC following flips the trail up
`Shifting ↑`	The POC has left the old value upward. Supportive	An early warning for the down trail
`Balance`	Value has not moved. The trail holds flat	The trail holds flat
`Shifting ↓`	An early warning for the up trail	The POC has left downward. Supportive
`Migrating ↓`	Value has left downward. A close below with the POC following flips the trail down	Value moving with the trail. Steps are likely

Add the WHY cell. A `probe` is a break without acceptance. On the default 126-bar trail fewer than one in twenty is followed by a flip within 20 bars, yet the trail's side then does worse than usual for 10 to 20 bars (section 16.2) MEASURED. "Accepted, flips at the close" means all three conditions hold on the live bar.

27.4 Matrix D · Asset class against evidence

Table 27.4 Asset class against evidence

Asset class	In the v9 audit (▲, grid) EXTERNAL	Accepted momentum, 2020-26 MEASURED	Accepted momentum, unseen markets MEASURED	Open interest
NSE stocks	Yes	+1.02	+0.84	Measured on NSE F&O EXTERNAL
US stocks	Yes	Not tested	Not tested	Not measured
Indices	Yes (world indices)	+0.98 (Indian indices)	Not tested	On index futures
Commodities	Yes	+0.47	Not tested	On futures
FX	Yes	−0.17 (INR pairs)	Not tested	None on spot
Crypto	Yes: the ▲ **failed** and is drawn gold	+0.29	−0.04	On perpetuals
Indian ETFs	Not a v9 class	+1.18	Not tested	None

In each era the ▲ was positive in four or five of the five non-crypto classes, so any one class may have missed an era. Figures are portfolio Sharpe ratios by class, net of costs.

CHAPTER 28

Conflicts and the decision tree

28.1 When readings disagree

1. **Data first.**
   Non-standard chart, warming, paused, stale or gold: there is no reading. Do nothing new.

2. **The horizon chooses the tool.**
   A trading book follows Accepted momentum. An investment book follows the grid's units. Do not let one book borrow the other's signal.

3. **Gold reduces size; it never sets direction.**
   Held rows, probes, quiet regimes, crowded books and short covering all mean the same thing: smaller, or later.

4. **Do not trade the untested parts.**
   The 20-bar trail, the ◆, R and H measured null or negative as signals. Read them as structure.

5. **Crypto is the exception.**
   Skip the capitulation plan there.

28.2 The decision tree

> **AFTER THIS CHAPTER YOU CAN**
> - go from a chart to an action through a fixed sequence of questions

1. **Is the bar closed, and is every layer I need readable?**
   No → do nothing yet. Yes → continue.

2. **Am I acting as a trader or as an investor?**
   Trader → 3. Investor → 7.

3 **Trader: what does the CHART cell say?**
LONG or SHORT → 4. FLAT · disagree → hold no position in this market; done.

4 **Is today the rebalance day?**
Yes → move to the target if it changed by more than 10% (section 19.1); done. No → 5.

5 **Discretionary only: was there a break of value with the trend at this close?**
Yes → optionally take the next part, within the target (T2). No → 6.

6 **A probe against my position?**
Yes → hold; optionally a step off (T4). Never exit on it alone, never add on it.

7 **Investor: which family, and is anything gold?**
Gold (held, quiet, settling, crypto capitulation) → wait. Otherwise → 8.

8 **Is it a Buy or Accumulate family?**
Yes → stage in by thirds (I3); add the second third on the ▲ in DOWN · cheap (I1). No → 9.

9 **PAID, STALLING or DISTRIBUTION?**
Trim toward 0.75, or exit toward 0.25 (I4). Otherwise hold, and rebalance on the calendar (I5).

PART X

Mastery

Practice

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CHAPTER 29

Drills

29.1 Reading drills

> **AFTER THIS CHAPTER YOU CAN**
>
> - read eight readouts without the book

1 Which family is this? Is the ▲'s condition present? What does the unshaded cell hold? Should an investor wait for PUSH ↑?

CVG `Buy 3.40` · WHY *sellers running* · *turning up* · PUSH ↓ · the pane tinted a deeper green.

2 What happened to the conviction tape, and what does an investor do?

CVG `Hold 1.31` · WHY *buyers pausing* · *held* (gold).

3 Name the regime. What is the system's position, and what does the probe say about longs?

AUCTION · STATE *Balance* · WHY *trail ↑* · *probe, not accepted* · CHART `FLAT · disagree`.

4 Which setup card is in context?

AUCTION · STATE *Migrating ↓* · WHY *trail ↓* · *below value* · CHART `SHORT 0.55`. The previous close was inside value.

5 What is the flow, and what should it change?

OI `Short covering` · WHY *exits* · *book 41 pct* · PUSH +1.2σ · the histogram's columns cast toward gold.

6 What does the OI layer contribute today?

OI `Warming · building its baseline`.

7 Which signals are paused? Is the grid readable?

A gold label at the pane's right edge: *SIGNALS PAUSED · value model building*.

8 Family? Why is it above 0.25 units? What does the investor do?

CVG `Exit 0.40` · WHY *sellers pausing · still richening*.

29.2 Context drills

9 Name the regime and what each of the trader and the investor does.

The weekly grid reads DISLOCATED. On the daily: CHART `SHORT 0.45`, CVG `Buy 2.90` TURNED.

10 Is the long riskier because the trail is so far away?

The panel's raw-share row reads QUIET. The trail is 25% below price. CHART `LONG 0.62`.

11 Does the daily trader act?

On a 15-minute chart the CHART cell turns FLAT at 11:30.

12 Does the capitulation plan apply?

BTC/USDT: CVG `Buy 3.70` DISLOCATED, the pane tinted gold, then a gold ▲.

29.3 Fakeout drills

13 Score the break. How often were breaks like it accepted within 20 bars?

Accepted momentum **off** (the 20-bar trail). A close 0.8 ATR below an up-trail, participation w = 2.1, the push gate −1, AUCTION Balance.

14 What matters now, and what is the likely outcome?

The same bar, but on the default 126-bar trail.

15 Exit? Add? What does the evidence say happens next?

You hold LONG 0.40 from the system. A gold × prints below the trail.

16 Is this a buy signal?

Two days later price closes back above the trail.

17 What is this, and what may a discretionary trader do?

CHART `LONG 0.38`. Yesterday's close was inside value; today's is above VAH.

18 Is this a better place to add?

CHART `LONG 0.38`. Yesterday's close was above VAH; today's is back inside value.

29.4 Sizing drills

19 Target weight and notional?

CHART `LONG 0.50`, annualised volatility 25%, slice ₹2,00,000.

20 Target weight and notional?

CHART `SHORT 0.30`, annualised volatility 6%, slice ₹2,00,000.

21 Target weight?

CHART `LONG 0.90`, annualised volatility 3.5%.

22 Weights, before any per-name cap?

Six names with units 4, 3, 1.5, 1, 0.75 and 0.25.

23 Do you trade at tomorrow's open?

It is not a rebalance day, and the CHART cell flips from LONG to SHORT.

24 Do you trade?

Rebalance day: the target moved from 0.40 to 0.43.

CHAPTER 30

Answers and checklists

30.1 Answers

1 DISLOCATED (DOWN · cheap), Buy · capitulation. *Sellers running* confirms the row; *turning up* is value REVERTING, the ▲'s condition. If value turned at this close, the ▲ prints on it. The deeper tint says the same. The unshaded cell holds 4.00. No: inside the capitulation cell the push's direction made no consistent difference, so a Buy there does not wait for a push ↑ (section 7.3).

2 The conviction tape has fallen below +30, but conviction's histogram has not confirmed a push down, so the UP row is HELD (section 7.1). The units sit between Hold and half-way to the FAINT row. The investor makes no adds and no changes. Held is caution, not direction.

3 Transition. The trail is up, momentum's S is not, and the system is flat. A probe below an up-trail is a caution on the long side: on the default trail, the side did worse than usual for 10–20 bars after probes (section 16.2). No long trade.

4 T2, the break of value with the trend (section 20.3). The system is short, and the close has left value in the trend's direction. The system trades on its rebalance; a discretionary trader may take the next part within the target.

5 Price is up while open interest falls: shorts are closing. Colouring it as self-limiting is a hypothesis {U}; pushes paid for by exits lagged other pushes in the v9 audit {X}. It changes size, never direction: a step smaller, or later.

6 Nothing. The feed is live, but it has fewer than ten accepted changes for its baseline. Skip the layer (section 9.2).

7 The ▲▼ (they need both level tapes) and the ◆, R and H. The grid is not readable while a level tape is not ready (section 10.2).

8 DISTRIBUTION (DOWN · rich). Both edges are unconfirmed, sellers pausing and value still richening, so the position sits half-way toward the middle and the units are above the cell's 0.25. The investor exits toward a token weight; the ▼ marked the entry (section 23.4).

9 A dislocation inside a downtrend: Matrix A's TURNED × AM SHORT, the classic conflict between horizons (section 27.1). The trader stays short by the system's rules. The investor builds in stages, with small tranches.

10 No. The quiet regime shuts the push gate, so the trail cannot step and its distance means nothing. The side and S carry the position. Stops come from structure (section 21.2), and the book's risk from the drawdown ladder (section 19.4).

11 No. The daily trader decides at the daily close. On a 15-minute chart the 126-bar trail and S's horizons are measured in 15-minute bars: a different and untested engine (section 14.3).

12 No. The capitulation reading failed on crypto, which is why its tint and ▲ are drawn gold (section 15.3).

13 Three of four conditions: depth ≥ 0.5 ATR, weight ≥ 1.5, and the push against the trail; migration toward the break is missing. On the 20-bar trail, breaks meeting three conditions were accepted about 70% of the time within 20 bars, against about 50% with none (section 16.3).

14 On the 126-bar trail only migration toward the break mattered, and here it is Balance. Fewer than one probe in twenty was accepted within 20 bars. Hard, heavy breaks were if anything less likely to flip it. Expect no flip, and treat the probe as a caution on longs.

15 Neither exit nor add. The system does not exit on a probe; a discretionary trader may take one step off (T4). After probes the trail's side did 0.05 to 0.09σ worse than usual over 10–20 bars (section 16.2).

16 No. The rejection entry, the first bar back after a probe, measured nothing and was negative in HA (section 16.4).

17 A break of value with the trend, the setup lab's S4: +0.06σ over 10 bars beyond the trend's average bar in all three sets, though it was not predicted. A discretionary trader may take the next part of the position, within the target (section 20.3).

18 No. A pullback into value carried no edge beyond the trend's average bar, and breakouts beat it everywhere (section 16.4).

19 0.50 × 10% ÷ 25% = 0.20. Notional = 0.20 × ₹2,00,000 = ₹40,000 long.

20 0.30 × 10% ÷ 6% = 0.50, short: −0.50 × ₹2,00,000 = −₹1,00,000. Low volatility gives large weights.

21 0.90 × 10% ÷ 3.5% = 2.57, capped at 2.

22 The sum is 10.5, so the weights are 38.1%, 28.6%, 14.3%, 9.5%, 7.1% and 2.4%. A per-name cap of about a tenth would redistribute the first two (section 24.1).

23 No. In the tested system every change waits for the weekly rebalance, flips included (section 19.1).

24 No. The change is 0.03, which is 7% of the new target, inside the 10% band.

30.2 Checklists

Every close (trader)

- [ ] Bar closed; every layer readable
- [ ] CHART cell on every market in the book
- [ ] Rebalance day? Targets that moved more than 10%
- [ ] Probes against positions noted; no exits on them
- [ ] Breaks of value with the trend noted (T2)
- [ ] Journal line written

Every week (investor)

- [ ] Weekly chart read first for each name
- [ ] CVG STATE, units and WHY at the week's close
- [ ] ▲▼ and tint changes since last week
- [ ] Weights = units ÷ sum, capped; move half the gap
- [ ] Trend filter applied, if used (I2)
- [ ] Clusters counted

Before any trade

- [ ] Which setup card? (name it)
- [ ] Context holds (regime, location)
- [ ] Size from the formula, rounded down
- [ ] Invalidation written
- [ ] Not a trap from chapter 17

Before going live

- [ ] Six months of paper trading at the test's costs
- [ ] Markets chosen and fixed; clusters capped
- [ ] Drawdown ladder and kill rule written down
- [ ] Book volatility target chosen (the tested 1.3–1.6%, or scaled)
- [ ] Start at a tenth of the intended size

PART XI

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CHAPTER 31

Alerts and settings

31.1 Alerts

Table 31.1 Alerts

Alert	How	Carries
Everything, once	Create an alert on Pragati with the condition "Any alert() function call"	▲, ▼, capitulation entered and left (setting 2), Accepted momentum's position changes or trail flips (setting 5, off by default), with ticker, interval, grid action and both tapes
One event	Choose a single alertcondition entry	▲ capitulation turn · ▼ distribution · ◆ long / short · R and H divergences

Alerts fire once, at the bar's close, in realtime only.

31.2 Every setting

> **AFTER THIS CHAPTER YOU CAN**
>
> - find any setting, its default and its range

All 80 inputs, in the order of the settings page, generated from the script itself. The defaults are the measured configuration; the engines' groups sit last because tuning them does not transfer between eras `EXTERNAL`. Each entry gives the first sentences of the setting's own tooltip.

Table 31.2 1 · View

Setting	Default	Range or options	What it does
View	Full	Full · Core	CORE is the instrument: histogram, trace, zones, markers and the ladder tapes; the profile, readout and gold cast stand down and are not computed. FULL (default) adds the price-chart context: the profile and the readout (open interest, the grid, the auction).
Palette	Auto	Auto · Light · Dark	AUTO follows the chart background. LIGHT and DARK force the palette.
Pane background	Grid state	Grid state · Declaration · Off	DISPLAY ONLY. A faint pane tint.

Table 31.3 2 · Signals · ▲ ▼ ◆

Setting	Default	Range or options	What it does
Signals ▲ ▼	On	—	THE ▲▼, read FROM THE GRID, where the v9 audit found the one robust edge. ▲ CAPITULATION TURN - the first closed bar on which the grid stands in DOWN · cheap (sellers in control across the ladder, value cheap past θ) with value momentum REVERTING. On 380 instruments in six classes, scored ...
Cooldown (bars)	10	0 to 200	Minimum bars between signals in the same direction - ▲ and long ◆ share one clock, ▼ and short ◆ the other. Without this, one choppy stretch produces a cluster of markers that all describe the same event.
Alert on capitulation state entry / exit	On	—	For alert() only: one alert with 'Any alert() function call' delivers every event, once at its bar's close, with ticker, interval, the grid's action and units, and both tapes - the ▲ and ▼, the trail's or Accepted momentum's changes if that alert is on, and, while this is on, the grid ENTERING ...
Continuation ◆	Off	—	OFF BY DEFAULT: on 380 instruments it was negative in both eras outside crypto (−0.021σ, −0.041σ over 10 bars), significantly on US stocks. Switch it on to see it; do not lean on it.
◆ · pullback window	6	2 to 50	For the ◆ only: how recently the histogram must have crossed to the wrong side for the move to count as a genuine pullback rather than an uninterrupted drift.
◆ · effort → result gate	On	—	The indicator's own thesis - is the move paid for - as a gate on the ◆: participation must NOT be absorbed on the signal bar (effort → result outside the bottom fifth of its history), so the resuming push is producing result. The ▲▼ do not read it: requiring absorption measured no better than ...

Table 31.4 3 · Trace and MTF tapes

Setting	Default	Range or options	What it does
Trace	Conviction × Value	Conviction × Value · Conviction · Value	WHAT THE PANE DRAWS. The histogram is always this trace minus its signal EMA.
MTF conviction ladder	Ladder down	Ladder down · Ladder up · Off	Which ladder of timeframes feeds the MTF CONVICTION tape (+82) and its MOMENTUM tape (+89). The value tapes have their own setting below.
Conviction momentum	Separate band	Separate band · Amalgamated · Off	MTF CONVICTION MOMENTUM - Dhṛti's momentum on the conviction ladder: the faster view of conviction minus the slower, in the tape's points. This sets how it is DRAWN, never what it reads.
MTF value ladder	Ladder up	Ladder up · Ladder down · Off	Which ladder feeds the MTF VALUE tape (−82) and its MOMENTUM tape (−89) - Samanvaya's MTF bias and momentum. LADDER UP (default, as in Samanvaya) - the chart and the standard frames above it, 4h the in-traday ceiling: 5m reads 4h · 1h · 30m · 15m · 5m; daily W · D; weekly M · W.
Value momentum	Separate band	Separate band · Amalgamated · Off	MTF VALUE MOMENTUM - Samanvaya's momentum on the value ladder: the faster view of value minus the slower; positive toward rich. This sets how it is DRAWN, never what it reads.

Table 31.5 4 · Pane display

Setting	Default	Range or options	What it does
Marker placement	Threshold rails	Threshold rails · On the oscillator	THRESHOLD RAILS stamps every event on a fixed level - divergences on the outer dashed lines, triggers on the inner dotted lines - with bearish above and bullish below, so the markers form readable rows. ON THE OSCILLATOR places each marker at the oscillator's value when the event fired.
Histogram	On	—	The trace minus its signal EMA - the pane's primary read. Off leaves the trace, rails and markers.
Histogram display gain	2.0	1.0 to 10.0	DISPLAY ONLY. Multiplies the plotted columns and the impulse band so the histogram (native range about ±20) is visible on a ±100 pane.
Histogram shading	Expansion + slope	Expansion + slope · Expansion only	DISPLAY ONLY. How each column's shade is decided.
Quiet-regime desaturation	On	—	DISPLAY ONLY. Adaptive scaling divides by 3σ of the raw share, so when that σ collapses a small imbalance reads large.
Impulse threshold band	Off	—	Draws the ± k·σ band - the impulse threshold the grid's push and the ◆ are judged against - over the histogram, at its display gain. Useful while calibrating k to an instrument; clutter afterwards.

Setting	Default	Range or options	What it does
Trace intensity by zone	On	—	DISPLAY ONLY. The trace grades with distance from zero: muted inside the inner rail, clearer past it, full past the outer; a blended trace also dims while its ingredients disagree or the basket settles.
Trace fill	On	—	DISPLAY ONLY. A faint fill between the trace and zero, green above and red below.

Table 31.6 5 · Acceptance trail · price chart

Setting	Default	Range or options	What it does
Acceptance trail	On	—	A stepped line on the PRICE chart: where value was last accepted. With Accepted momentum on (default) it draws that tested engine - side as hue, size as intensity; off, the Lookback-window trail, structure that measured null as a trade.
Accepted momentum · the tested engine	On	—	ON (default): the trail becomes ACCEPTED MOMENTUM, the one trail engine here with a tested edge (docs/trail-lab.md). SIDE - this trail on a 126-bar (six-month) window.
Value ribbon	On	—	The rolling value area behind the trail: where 70% of the window's participation traded (capped as the conviction engine caps it), in its own slate - not the session profile's blue - with its POC dotted. The last 1,500 closed bars.
Probe marks	On	—	A gold × at the close of the FIRST bar of each probe - a close beyond the trail the market had not accepted (the POC not following, or value not migrated). A probe lasts about four bars; marking every bar of it turned a quarter of the chart gold.
Alert on trail flips	Off	—	Adds to the one 'Any alert() function call' alert, with the chart's state. With Accepted momentum on: each change of its position - LONG, SHORT or FLAT - with the size.

Table 31.7 6 · Readout · price chart

Setting	Default	Range or options	What it does
Readout position	Bottom left	Bottom left · Bottom centre · Bottom right · Middle left · Top centre · Top right	Where the readout - the OI, CVG and AUCTION rows, one table - sits on the price chart. Default BOTTOM LEFT: the profile's histogram and level tags take the right, the legend the top left.
Open interest row	On	—	The OI row of the readout: the book's character (STATE), who is doing it and how full the book is (WHY), the change in σ (PUSH), how long it has held (SPAN). A feed that cannot be measured says so.

Setting	Default	Range or options	What it does
OI · gold cast on histogram (exits)	On	—	ON BY DEFAULT - the one place OI reaches the pane, as COLOUR only. A column is cast toward gold when the push on its side over the last signal-length bars was paid for more by EXITS than by new positions (covering outnumbering long buildup above zero, unwinding outnumbering short buildup below), ...
Conviction-Value Grid row	On	—	The CVG row of the readout - Pragyam's Conviction-Value Grid, read for this chart. STATE - the family's action (Buy, Accumulate, Hold, Wait, Trim, Exit) and its units.
Auction row in the readout	On	—	The AUCTION row of the readout. STATE - value's migration against the window N bars ago (MIGRATING, SHIFTING, BALANCE).

Table 31.8 7 · Volume profile · price chart

Setting	Default	Range or options	What it does
Volume profile	On	—	ON by default. Off removes every chart-overlay element AND stops the profile being computed - accumulation, value-area walk and naked-POC ledger all stand down.
Build from intrabar data	On	—	request.security_lower_tf() distributes volume by ACTUAL traded price. More accurate, slower, plan-dependent history.
Volume-weighted	On	—	Off degrades the profile to a time-at-price (TPO) distribution — for spot FX / synthetic volume.
Bin grid	ATR adaptive	ATR adaptive · Fixed ticks	
Bins per ATR · ATR mode	4.0	1.0 to 20.0	Bin width = ATR / this. Frozen at session start so keys stay consistent within a profile.
Bin size (ticks) · fixed mode	20	1 to 5000	
Value area fraction	0.70	0.50 to 0.95	
Histogram	On	—	
Histogram style	Monochrome	Monochrome · Delta split	MONOCHROME (default): one box per row - a bright neutral POC row inside blue value-area rows, grey tails; hue marks identity, never direction. DELTA SPLIT: each row splits into green (inferred buy) and red (inferred sell) volume, the POC row outlined.
Histogram width (bars)	30	8 to 150	
HVN / LVN markers	On	—	
Value area (VAH / VAL)	On	—	
POC line	On	—	

Setting	Default	Range or options	What it does
Developing level trails	On	—	
Prior session value	On	—	Last session's pPOC, pVAH and pVAL — three dotted lines, drawn only across the session in front of you. These are the reference the market is currently accepting or rejecting, which is why they are not stacked back through history.
Naked POC (untouched)	On	—	
Naked POC · max kept	5	1 to 20	How many untouched POCs the ledger carries; the oldest is dropped when it is full. Twelve dotted lines on a long chart become the chart; five shows the unfinished business above and below.
Naked POC · max age (bars)	250	50 to 1000	A naked POC older than this is dropped whether or not price ever traded through it, and no line starts further back than this. A point of control from four hundred bars ago describes a market that no longer exists.
Session VWAP	On	—	
Initial Balance (IBH / IBL)	On	—	The session's first span of bars (below), intraday charts only: on daily and weekly bars the span covers the whole session, so IBH / IBL would only restate its high and low.
Initial Balance span (bars)	12	3 to 60	

Table 31.9 8 · Information panel

Setting	Default	Range or options	What it does
Information panel	Off	—	The oscillator's own panel, in the indicator pane. Off removes it entirely.
Detail	Detailed	Compact · Detailed	COMPACT is the headline, the signal state, both tapes, the impulse and the trace's ingredients. DETAILED adds the raw share and its σ, this bar's conviction, the participation weight, and — when the profile is on — its data mode and where price sits in it.

Table 31.10 9 · Engine · conviction and zones

Setting	Default	Range or options	What it does
Lookback	20	3 to 200	How many bars of effort the oscillator accounts for. This is the horizon of the whole instrument.
Smoothing	3	1 to 20	Final EMA on the oscillator. Its real job is the divergence engine: pivots detected on a jagged series produce spurious highs and lows and therefore spurious divergences. 1 disables it.
Signal EMA	9	1 to 50	The reference line. The histogram is the gap between the trace and this.

Setting	Default	Range or options	What it does
Impulse threshold · k (σ of histogram)	0.5	0.0 to 3.0	The histogram's magnitude gate: a column is an IMPULSE once past ± k standard deviations of its own history. It does NOT gate the grid - rows move on any push, impulse or not.
Normalization window	200	30 to 1000	Sample used for the scaling σ and for the trigger's magnitude threshold. Longer is stabler and slower to acknowledge that the instrument has changed character.
Participation	Auto	Auto · Volume · True range · Off	What weights each bar's conviction. AUTO - volume where the symbol has it, relative true range where not, so index spot needs no setup.
Denominator	Agreement	Agreement · Effort	What the signed, participation-weighted result is divided by. AGREEMENT (default; the only form the evidence was measured on): Σ|c|·w - how consistently bars resolve one way.
Participation baseline	20	5 to 200	Averaging length for the volume or range baseline. Each bar is weighted by its participation relative to this.
Participation cap	3.0	1.0 to 10.0	Ceiling on the participation weight. Expiry days, index rebalances and block prints generate volume many multiples of normal.
Inner zone	30.0	5.0 to 95.0	Where the window's effort has become heavily one-sided. Under adaptive scaling this is occupied roughly a third of the time.
Outer zone	60.0	10.0 to 99.0	Near-uniform conviction, roughly four percent occupancy under adaptive scaling. Trends live here and so do their endings.

Table 31.11 10 · Engine · value model (Samanvaya)

Setting	Default	Range or options	What it does
Macro hedge	Auto (measured)	Auto (measured) · Off · Full	How much of the macro hedge the RV leg applies. AUTO scales it by the hedge's rolling out-of-sample R^2 (across 24 targets it beat a full hedge on mean and median).
Model clock	Daily (works on every timeframe)	Daily (works on every timeframe) · Weekly (stronger · needs daily+ charts)	Clock on which the RV leg's driver selection and hedge ratios are estimated, then applied to the chart. DAILY works everywhere.
Leg mix · RV ← → breadth	0.5	0.0 to 1.0	Weight on the RELATIVE VALUE leg: 0.5 is the equal blend Samanvaya measured best (+0.0691 against +0.0637 for RV alone), 1.0 RV alone, 0.0 breadth alone. On value Ladder down the rungs are breadth only; the chart rung always uses the mix.

Setting	Default	Range or options	What it does
θ · value threshold	1.5	0.5 to 4.0	Samanvaya's entry threshold θ, in |z|: where value counts as CHEAP or RICH. It sets the grid's columns - so the ▲ (down · cheap past θ) and ▼ (sellers holding a price rich past θ) - the value tape's level for the ◆ and H, the Value and blended traces' inner rail, and the knee of the value tape's ...
Add a custom macro driver	Off	—	Adds your own symbol as a ninth candidate factor in the RV leg. Never the target's own volatility index - it would hedge the target against itself.
Custom driver symbol	—	—	Leave empty unless you have a specific economic reason.

Table 31.12 11 · Divergence · R H (measured null)

Setting	Default	Range or options	What it does
Regular divergence R	Off	—	REGULAR DIVERGENCE - the move is not confirmed by the chart OR the ladder, at a price every horizon calls stretched. R BEARISH (bullish mirrors), at the confirming pivot: CHART price a higher high, conviction a lower high (conviction's own pivots; with the zone gate, the earlier high ...
Hidden divergence H	Off	—	HIDDEN DIVERGENCE - a pullback where control never left. H BEARISH (bullish mirrors), at the confirming pivot: CHART price a lower high, conviction a higher high (its own pivots) CONVICTION the conviction tape past the inner zone on the sellers' side VALUE the value tape NOT ...
Pivot left	5	1 to 30	Bars of confirmation on the left of a pivot.
Pivot right	5	1 to 30	Bars of confirmation on the right of a pivot. THIS IS THE LAG.
Min pivot separation	5	2 to 100	Two pivots closer than this are the same swing and their comparison is meaningless.
Max pivot separation	60	10 to 400	Two pivots further apart than this belong to different market episodes. Comparing them produces impressive-looking lines that mean nothing.
Zone gate on regular divergence	On	—	Only accept a regular bearish divergence whose earlier pivot formed above the inner zone (bullish: below). Equivocal: it improved quality on gold, silver and crude and slightly reduced it on eight held-out instruments; on by default for the primary futures.
Draw divergence connectors	On	—	The line joining a divergence's two pivots - in the pane on the Conviction trace; on the price chart on the others (the drawn trace is not the series the pivots were found on). Off keeps only the rail markers.

CHAPTER 32

What is measured

32.1 What is measured, and what is not

Table 32.1 What is measured, and what is not

Component	Grade	Result
Accepted momentum	MEASURED	Holdout Sharpe +0.79 (411 unseen markets) and +1.21 (2020-26), net of costs; timing t 4.1 and 2.2; did not beat plain momentum
20-bar acceptance trail as a trade	MEASURED	No timing skill: what it earned was drift. Structure only
20-bar trail, long only (an investment filter)	MEASURED	Drawdown −35% against −73.5%, growth 13.5% against 24.4% a year; other trend filters made the same trade
▲ capitulation turn and state	EXTERNAL	Positive in all three eras, daily and weekly, outside crypto; about +0.05σ over 10 to 20 bars
▼ distribution state	EXTERNAL	Followed by underperformance in every era
Grid units	EXTERNAL	Positive in all three eras outside crypto
Declaration (held ▲▼)	EXTERNAL	No edge
◆, R, H	EXTERNAL	Negative, about zero, negative or mixed
Open interest	EXTERNAL	Short-covering rallies lagged; builds did not predict continuation
Probes, acceptance and entries around value (setup lab)	MEASURED	A probe weakens the trend's next weeks; on the 20-bar trail deep and heavy breaks are accepted more often; breakouts from value beat pullbacks into it (section 32.2)
Accepted momentum as an investor's filter (setup lab I1)	MEASURED	Drawdown roughly halved, return roughly halved; Sharpe lower than buy-and-hold in 2020-26
Session profile, HVN / LVN, naked POCs, VWAP, IB	UNMEASURED	Reference structure
Every combination in Part IX	UNMEASURED	Reasoning from the measured parts

Known limits: the Accepted momentum test is mostly Indian markets plus crypto, and its NSE stocks are today's listings (survivorship). The v9 audit's figures come from a separate

repository.

32.2 The setup lab's verdicts

Table 32.2 Pre-registered in docs/setup-lab.md, run once on 119 markets to 2019, the same markets in 2020–26, and 411 unseen markets

Test	Question	Verdict	Where
F1	Is a probe a fakeout?	The opposite: the trend's side did worse after probes, on both trails MEASURED	section 16.2
F2a–d (20-bar)	What gets accepted?	Depth and weight confirmed; push and migration consistent MEASURED	section 16.3
F2a–d (126-bar)	What gets accepted?	Migration consistent; depth, weight and push not supported MEASURED	section 16.3
S1	Enter on the rejection	Not supported MEASURED	section 16.4
S2	Enter on the acceptance	Consistent MEASURED	section 20.4
S3	Buy the pullback into value	Not supported MEASURED	section 16.4
S4	Buy the breakout from value	Positive with t ≥ 2 in all three sets at 10 bars; not predicted MEASURED	section 20.3
S5	Migration in a trend	Consistent MEASURED	section 8.1
I1	An investor's trend filter	Not confirmed: drawdown halved, Sharpe lower in 2020–26 MEASURED	section 23.2

Quick reference card

Every close

- Gold anywhere? Smaller, or later.
- Trader: read the AUCTION row's CHART cell; act at the next open.
- Investor: read the CVG row's STATE at the week's last close.

Trader

- LONG or SHORT only when trail and momentum agree.
- Weight = size × 10% ÷ volatility, cap 2.
- Many markets, weekly rebalance.
- Paper first, then a tenth.

Setups

- T1 Accepted momentum is the only system.
- T2 break of value: within the target.
- T4 probe against you: hold, never add.
- I2 trend filter: about half the drawdown, half the return.

Investor

- Weight = units ÷ total units.
- Stage into DOWN · cheap; add on the ▲.
- Trim in PAID and STALLING; exit in DISTRIBUTION.
- Not on crypto.

Breaks

- A probe is a caution on the trend, not a fakeout to fade.
- 126-bar trail: only migration toward a break makes acceptance likely.
- 20-bar trail: depth ≥ 0.5 ATR and weight ≥ 1.5 raise acceptance.
- With the trend, act on the break of value, not the pullback into it.

Never

- Trade the forming bar.
- Read gold as a direction.
- Lean on the ◆, R, H or the 20-bar trail.
- Use non-standard candles.

Glossary

Acceptance	A close beyond the trail with the window's POC beyond it and value migrated that way. Only acceptance flips the trail.
Build-up · covering · unwinding	Open interest rising with the price move (new positions), or falling (positions closing).
Campaign	A run of the traded weight on one side, from the rebalance that opens it to the one that closes it.
Capitulation	Sellers in control across the ladder at a price cheap past θ: the DOWN · cheap cell.
Conviction	The share of the window's participation-weighted effort that became net movement.
Family	One of the grid's nine cells, named for its action.
Flip	The trail changing side on acceptance: a close beyond it, the POC beyond it, and value migrated that way.
Held	A grid row kept in place because no push confirmed the tape's move.
HVN · LVN	High- and low-volume nodes: local peaks and troughs in the session profile.
IBH · IBL	The initial balance: the high and low of the session's first bars (12 by default), intraday only.
Impulse	A histogram column past k standard deviations of its history (↑ ↑).
Kill rule	Stop the book if its drawdown passes 1.5 times the backtest's worst, or its rolling 12-month Sharpe stays below zero for two quarters.
Ladder down / up	Reading the timeframes below the chart (inside each bar) or above it.
Migration	This window's value area against the one six months earlier: Migrating (no overlap), Shifting (only the POC left), Balance.
Naked POC	A prior session's POC that price has not traded back through.

Phase	A momentum's standing side against its level tape's edge: running or pausing for conviction, reverting or widening for value.
POC · VAH · VAL	The most-traded price; the top and bottom of the 70% value area.
Probe	A close beyond the trail without acceptance.
Probe onset	A probe's first bar, the one marked with a gold ×.
Push gate	Conviction's histogram on the move's side, not turning, not quiet.
Quiet regime	Raw volatility in the bottom fifth of its history: readings are magnified.
Rebalance band	The tested rule: on a rebalance day a market's weight changes only if its target moved more than 10% of itself.
S	Accepted momentum's score, −1 to +1, from the 1-, 3-, 6- and 12-month moves.
Setup card	The fixed form every setup in this book takes: context, trigger, entry, size, stop, exit, evidence and failure.
Sharpe	Return per unit of volatility, yearly. t is how far from zero it is, in standard errors.
Slice	A market's share of capital in an equal-risk book: the capital divided by the number of markets.
Structural bar	An OI change of more than half the book, or beyond 3σ and over a fifth of it: an expiry or roll, kept out of the statistics.
θ (theta)	Samanvaya's value threshold, ±43 on the pane at the default 1.5.
Transition	The regime in which the trail and momentum disagree, so Accepted momentum reads FLAT · disagree.
Units	The relative weight a family gives a name in a portfolio.
VWAP · TWAP	The session's volume-weighted average price; time-weighted when the symbol has no usable volume.

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